

Through the Looking Glass

A discussion of the art of microscopy

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Outline

1. A brief discussion of the historical development of microscopy (10 minutes)
2. A discussion of the workings of light microscopes (10 minutes)
3. A discussion of the Airy disk and the diffraction limit (5 minutes)
4. Improving the diffraction limit using electrons instead of light (5 minutes)
5. A discussion of the workings of electron microscopes (10 minutes)
6. Some sample micrographs (5 minutes)
7. Questions (10 minutes)



The historical development of microscopy

The Nimrud (Layard) lens
(8th century BC)



https://en.wikipedia.org/wiki/Nimrud_lens

The Nimrud lens, also referred to as the Laynard lens, is a piece of rock crystal dated back to the 8th century BC that was found by Austen Henry Layard at the Assyrian palace of Nimrud in what is now modern-day Iraq.

Historians believe that it may have been used as a magnifying glass or as a burning-glass to start fires by concentrating sunlight.

The historical development of microscopy

The Reading Stone
(~1000 AD)



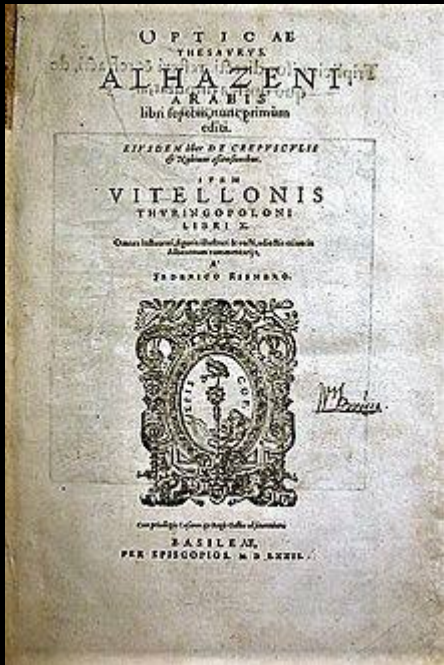
https://en.wikipedia.org/wiki/Reading_stone

A reading stone is an approximately hemispherical lens that can be placed on top of text to magnify the letters so that people can read them more easily.

The invention of reading stone is often credited to Abbas ibn Firnas in the 9th century AD; however, the regular use of reading stones began around 1000 AD.

The historical development of microscopy

The Book of Optics (~1021 AD)



The Book of Optics is a seven-volume treatise on optics and other fields of study composed by the medieval Arab scholar Ibn al-Haytham, known in the West as Alhazen or Alhacen.

The book extensively affected the development of optics (the underlying field of microscopy), physics, and mathematics in Europe between the 13th and 17 centuries AD.

https://en.wikipedia.org/wiki/Book_of_Optics

The historical development of microscopy

The Eyeglasses
(1284 AD)



Eyeglasses were invented by Salvino D'Armati in Italy during the 13th century AD.

It all started with the invention of two convex lenses placed in a wooden setting, with a shaft held together with a rivet.

<https://www.sciencelearn.org.nz/resources/1692-history-of-microscopy-timeline>

The historical development of microscopy

Anton van
Leeuwenhoek
(1632 – 1723)



Van Leeuwenhoek is best known for his pioneering work in microscopy, as he was one of the first people to make simple single-lens microscopes by grinding his own lenses.

He is commonly known as “the Father of Microbiology,” and is often considered the inventor of the first microscopy (though this is not confirmed).

https://en.wikipedia.org/wiki/Antonie_van_Leeuwenhoek

The historical development of microscopy

Zacharias Janssen
(1585 – 1632)



Janssen is associated with the invention of the first optical telescope and/or the first truly compound microscope that used two lenses, but these claims (made 20 years after his death) may be fabrications put forward by his son.

https://en.wikipedia.org/wiki/Zacharias_Janssen

The historical development of microscopy

The Transmission
Electron Microscope
(1931)



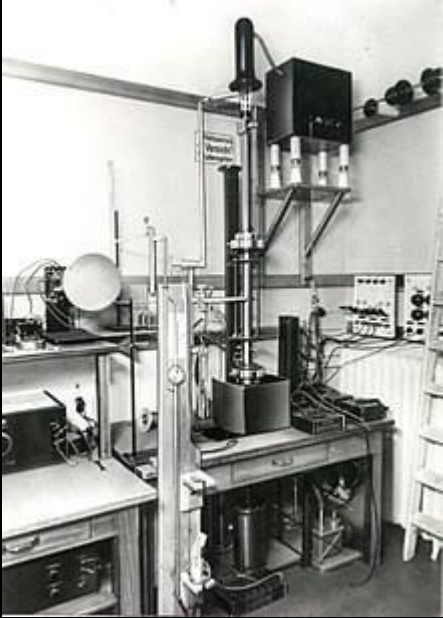
Ernst Ruska and Max Knoll designed and built the first transmission electron microscope (TEM), based on an idea of Leo Szilard.

The electron microscope using electrons, not light, to view and object. Modern TEMs can visualize objects on the length scales of atoms (sub-nanometer resolution).

https://en.wikipedia.org/wiki/Transmission_electron_microscopy

The historical development of microscopy

The Scanning Electron Microscope (1937)

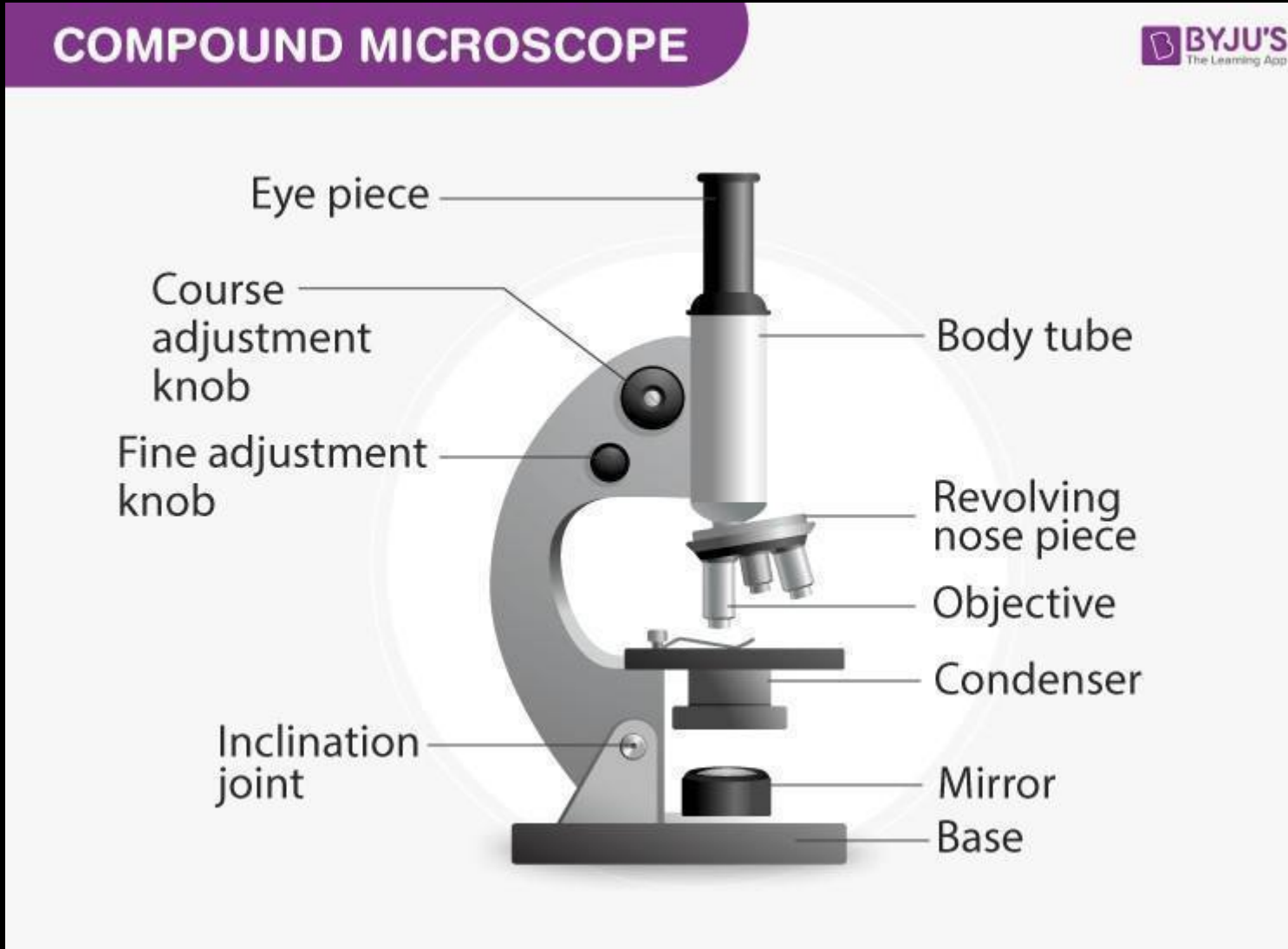


Manfred von Ardenne invented a microscope in 1937 that achieved a high resolution image by scanning a finely focused electron beam, and the scanning electron microscope (SEM) was born.

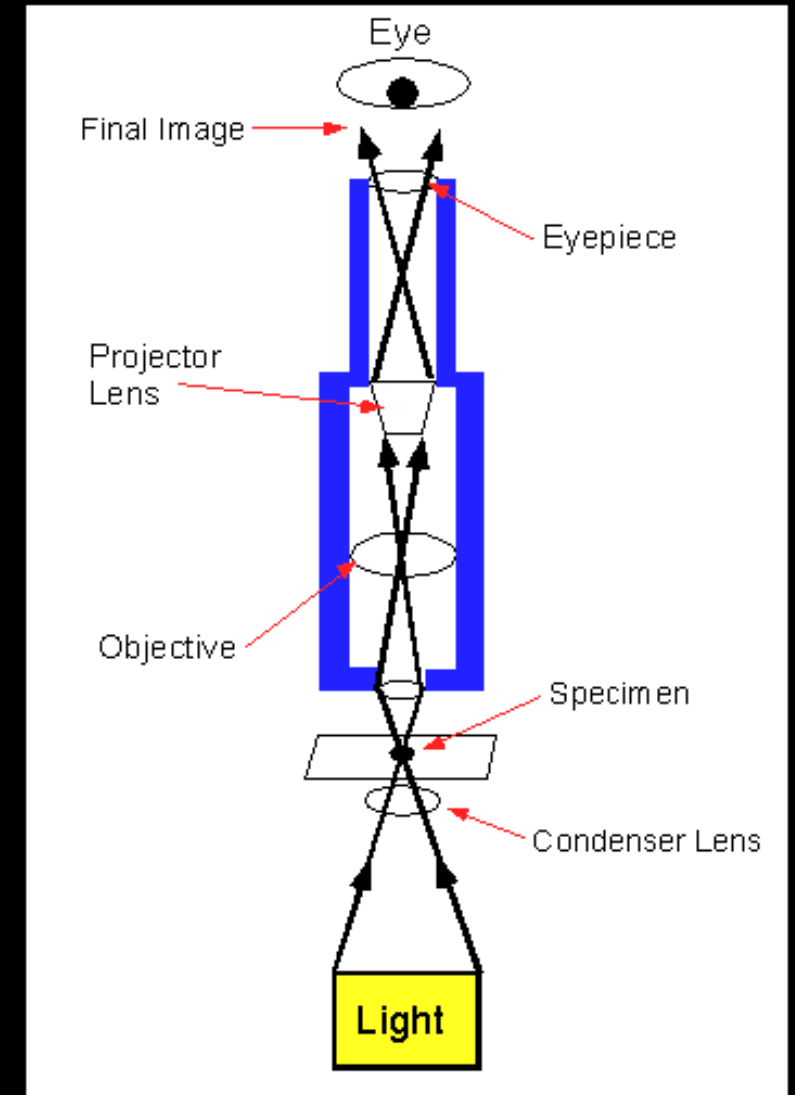
While SEMs cannot achieve the resolution of most TEMs, they are more akin to a reflection optical microscope and can see in three dimensions, where TEMs, by their nature, can only see in two dimensions.

https://en.wikipedia.org/wiki/Scanning_electron_microscope

The Workings of the Compound Light Microscope



<https://byjus.com/physics/compound-microscope/>



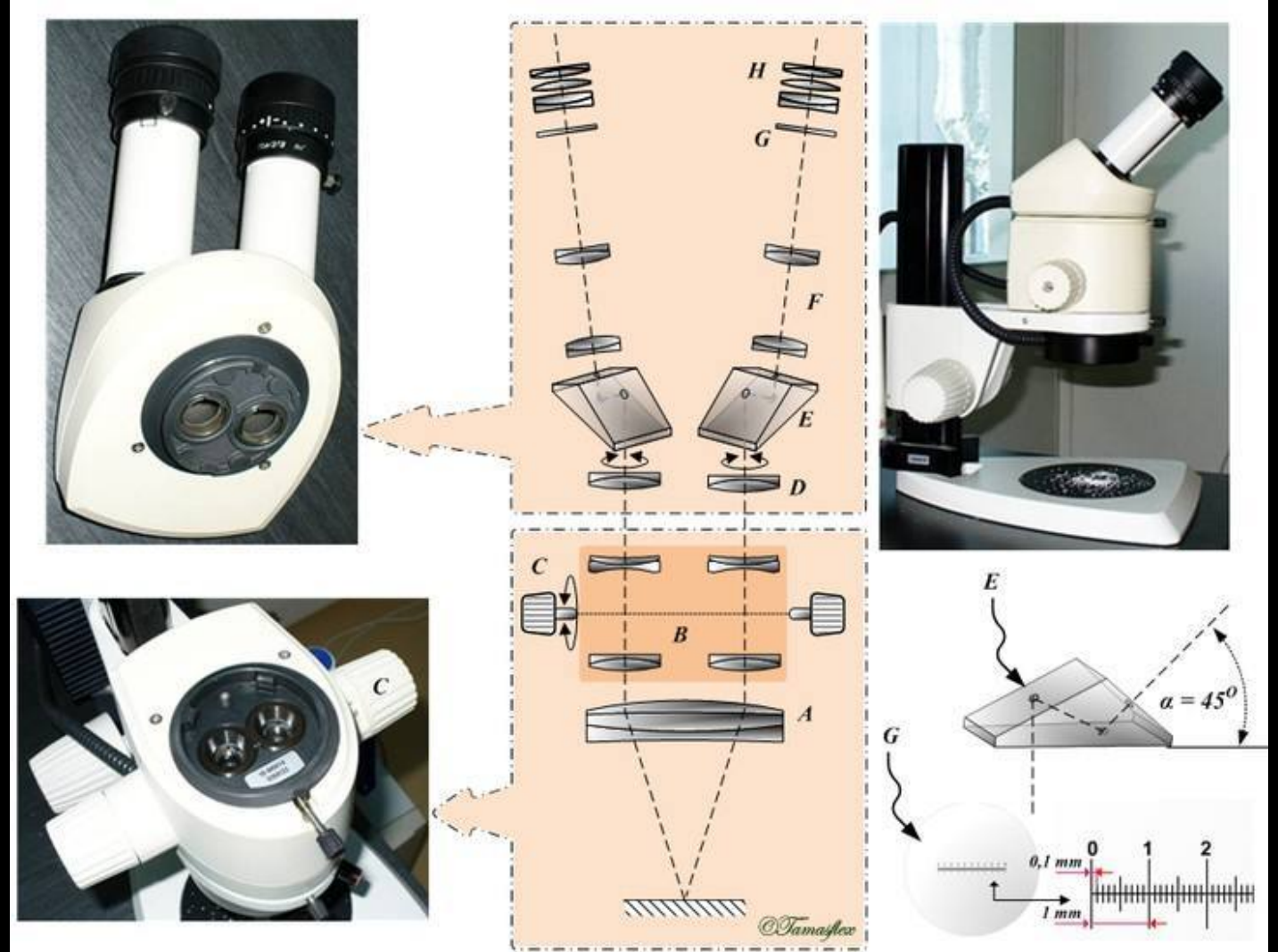
<https://www.cas.miamioh.edu/mbi-ws/microscopes/compoundscope.html>

image credit: Gary
DeLong
(www.fineartamerica.com)



Cross-section of a Pteris (fern) Rhizome at 100× magnification

The Workings of the Stereo Light Microscopy



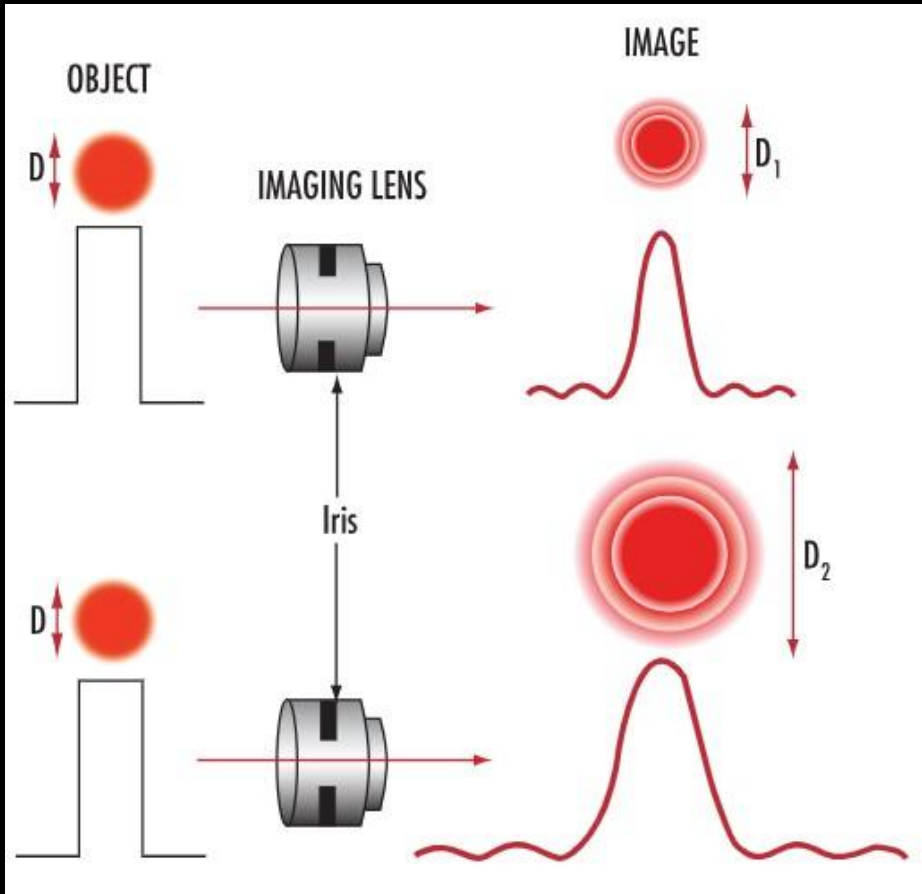
CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons



A common wasp (*Vespa vulgaris*) imaged using a stereo microscope

The Airy Disk and the Diffraction Limit

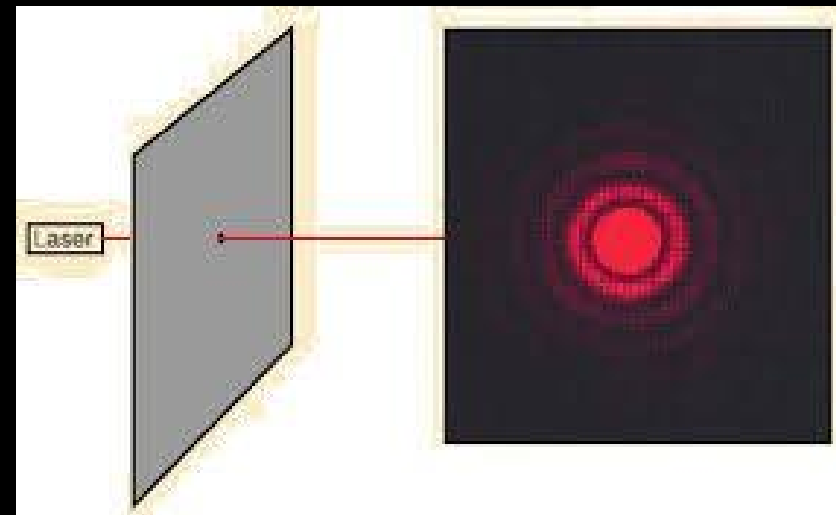
The Airy Disk



<https://www.edmundoptics.com/>

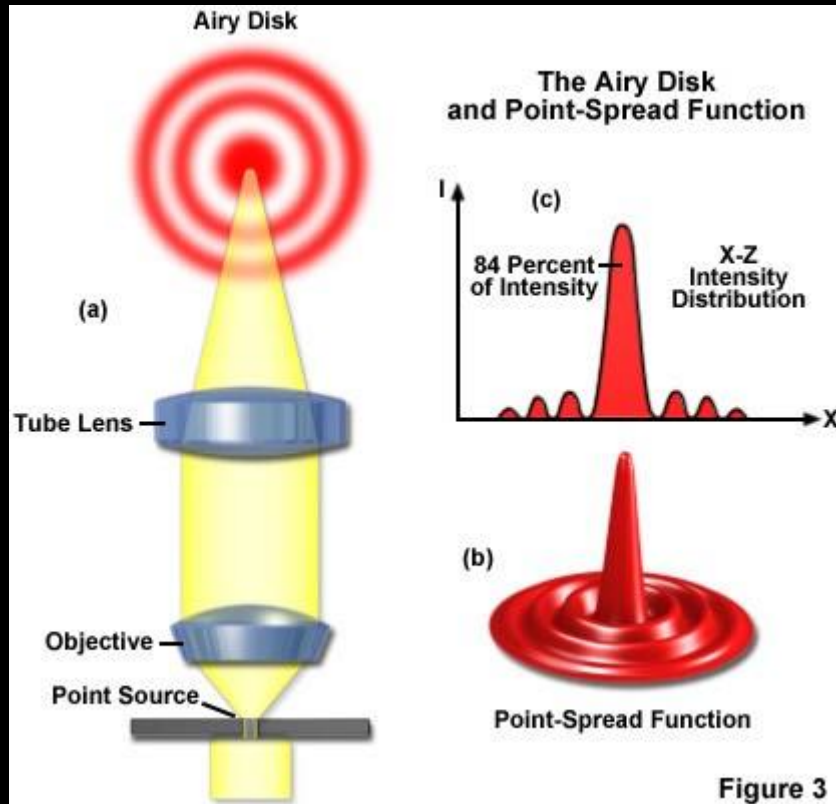
When light passes through any size aperture, diffraction occurs; the resulting diffraction pattern is called the **Airy disk** (for a circular aperture).

The diameter of the Airy disk is directly proportional to the wavelength (λ) of the illuminating light, and inversely proportional to the diameter of the aperture.

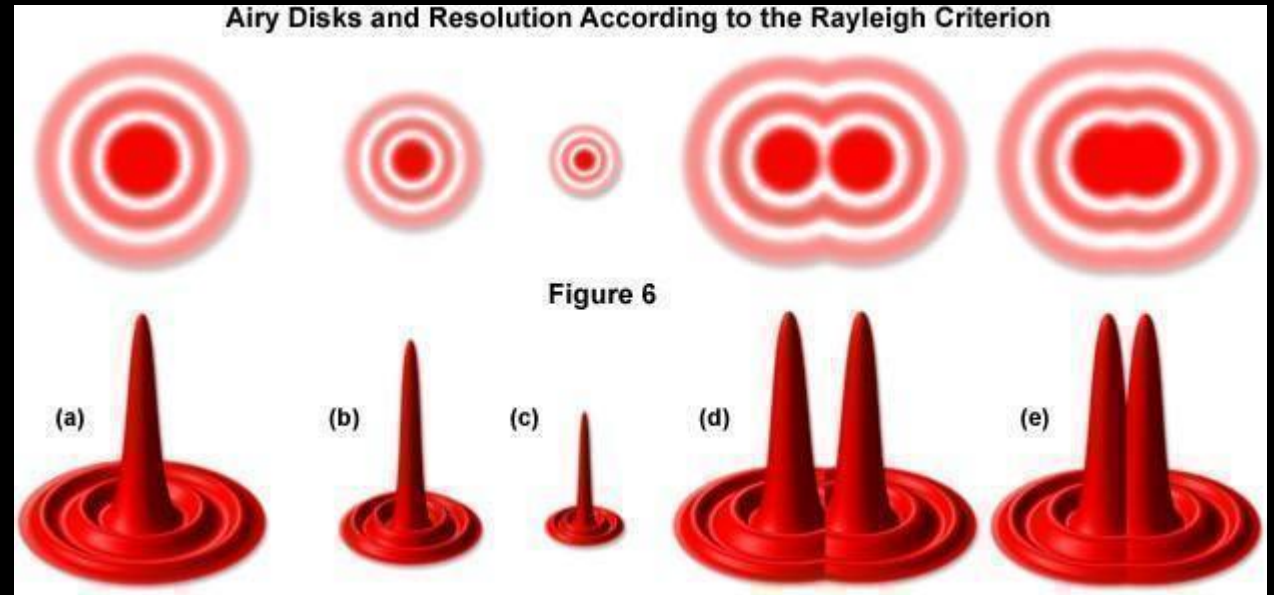


<http://hyperphysics.phy-astr.gsu.edu/hbase/phyopt/cirapp2.html>

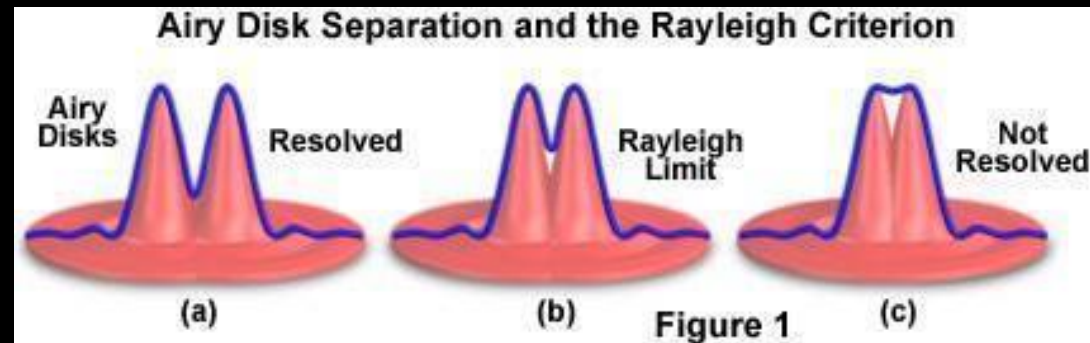
The Airy Disk and the Diffraction Limit



<https://zeiss-campus.magnet.fsu.edu/articles/basics/resolution.html>



<https://zeiss-campus.magnet.fsu.edu/print/basics/imageformation-print.html>



<https://www.olympus-lifescience.com/en/microscope-resource/primer/digitalimaging/deconvolution/deconresolution/>

Improving the Diffraction Limit Using Electrons

Louis de Broglie
(1892 – 1987)



https://en.wikipedia.org/wiki/Louis_de_Broglie

In his 1924 Ph.D. dissertation, de Broglie postulated the wave nature of electrons and suggested that all matter has wave-like properties. This concept is known as the de Broglie hypothesis, but is more commonly referred to as wave-particle duality.



<https://journals.aps.org/pr/abstract/10.1103/PhysRev.30.705>



https://en.wikipedia.org/wiki/Nobel_Prize

Improving the Diffraction Limit Using Electrons

Electrons have wave-like properties

The de Broglie Hypothesis

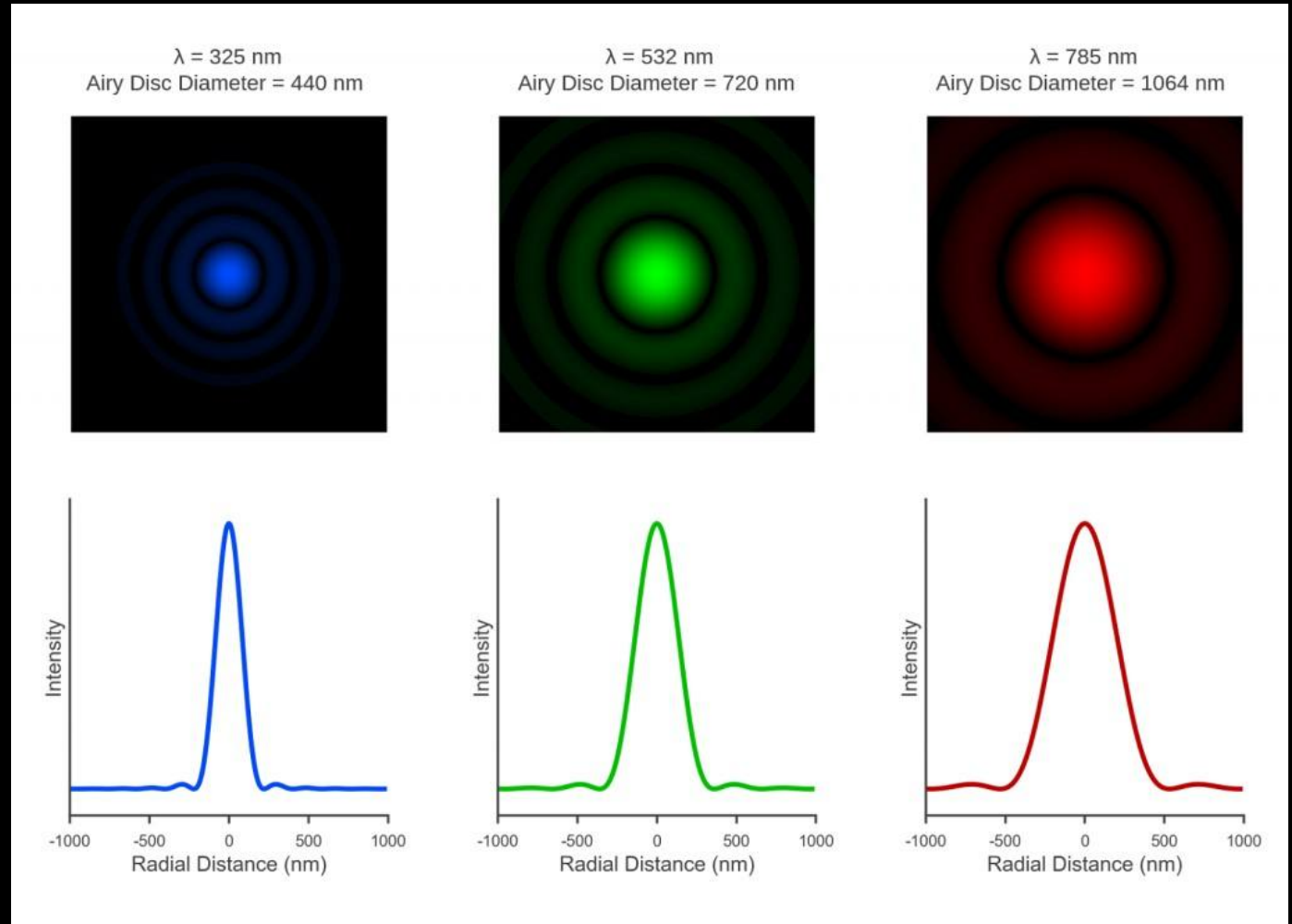
$$\lambda = \frac{h}{mv}$$

h — Planck's constant

m — the mass of the particle

v — the speed of the particle

The Airy disks produced by sending a laser through a high-magnification optical objective lens (100x; 0.9 NA)



<https://www.edinst.com/blog/laser-spot-size-in-a-microscope/>

Improving the Diffraction Limit Using Electrons

Electrons have
wave-like properties

The de Broglie

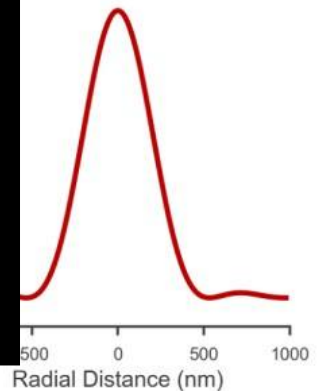
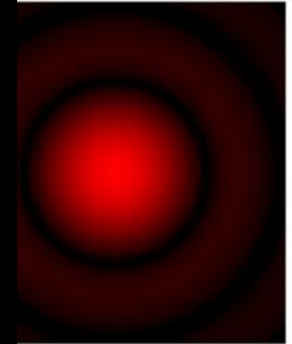
$$\lambda =$$

h — Planck's constant
 m — the mass of the electron
 v — the speed of the electron

Can we use electrons,
instead of light, for
microscopy?

through a high-
numerical aperture (NA)

$\lambda = 785 \text{ nm}$
Microscope Diameter = 1064 nm



Radial Distance (nm)

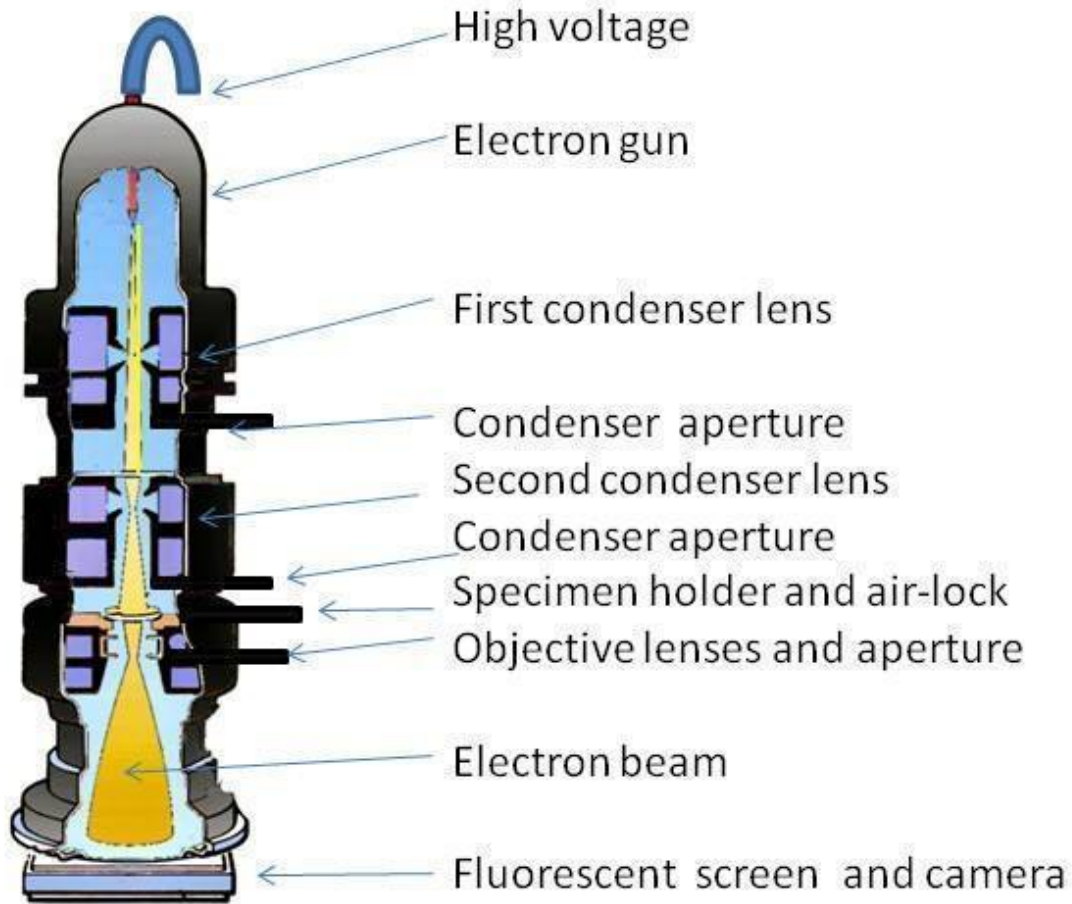
Radial Distance (nm)

Radial Distance (nm)

<https://www.edinst.com/blog/laser-spot-size-in-a-microscope/>

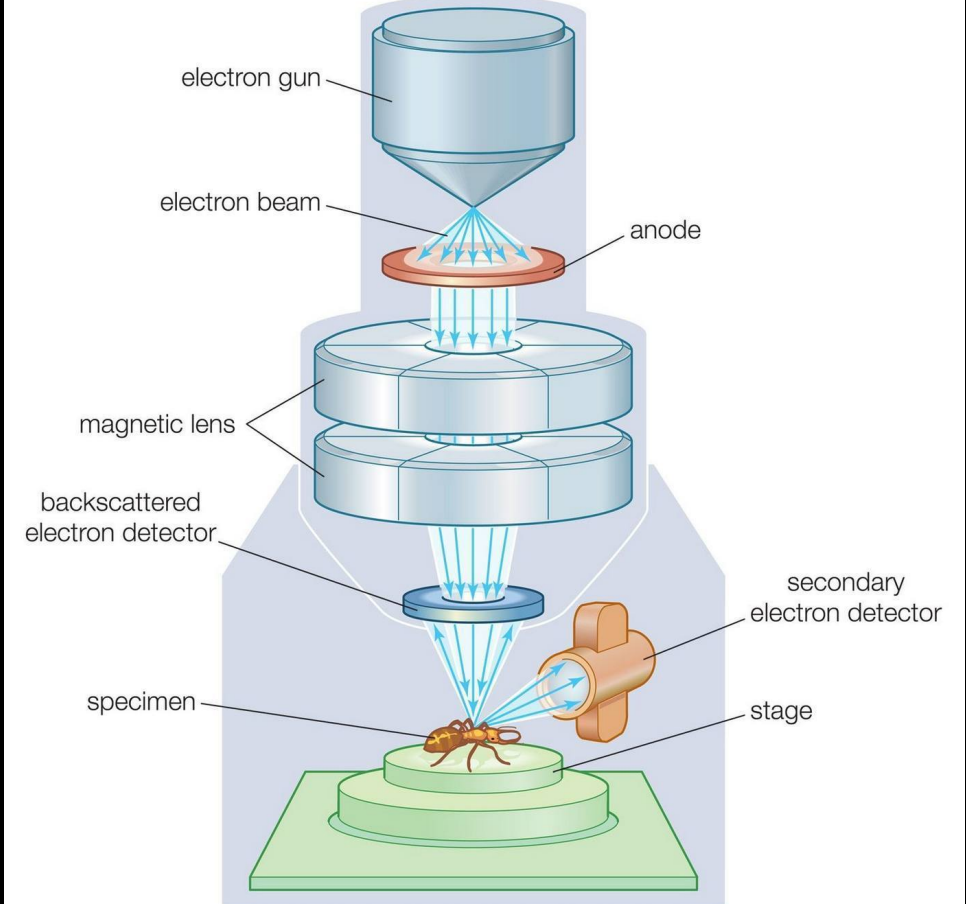
The answer is a resounding yes!!!

The Transmission Electron Microscope (TEM)



<https://www.ccber.ucsb.edu/ucsb-natural-history-collections-botanical-plant-anatomy/transmission-electron-microscope>

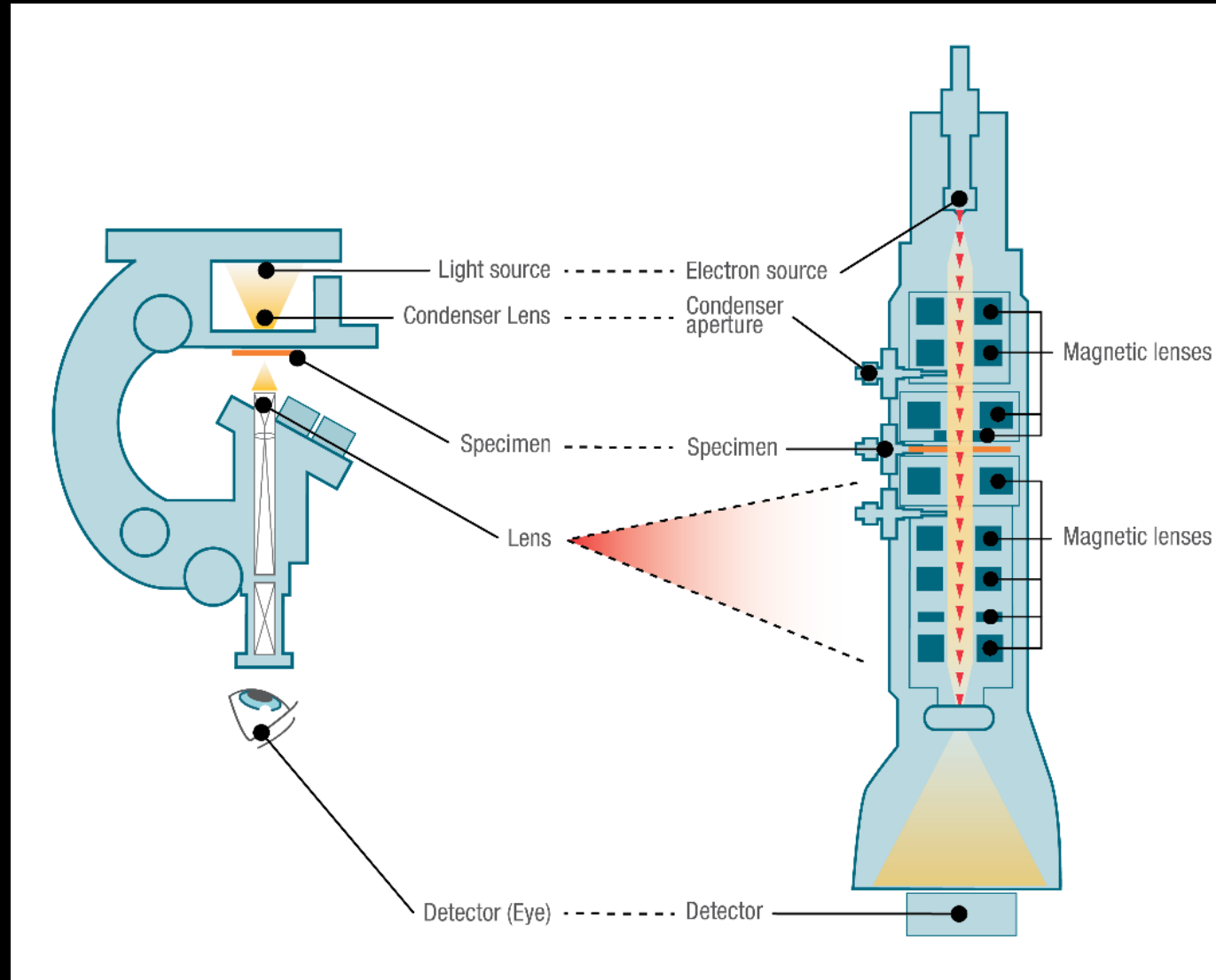
The Scanning Electron Microscope (SEM)



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<https://www.britannica.com/technology/scanning-electron-microscope>

A TEM operates on the same principle as a compound light microscope

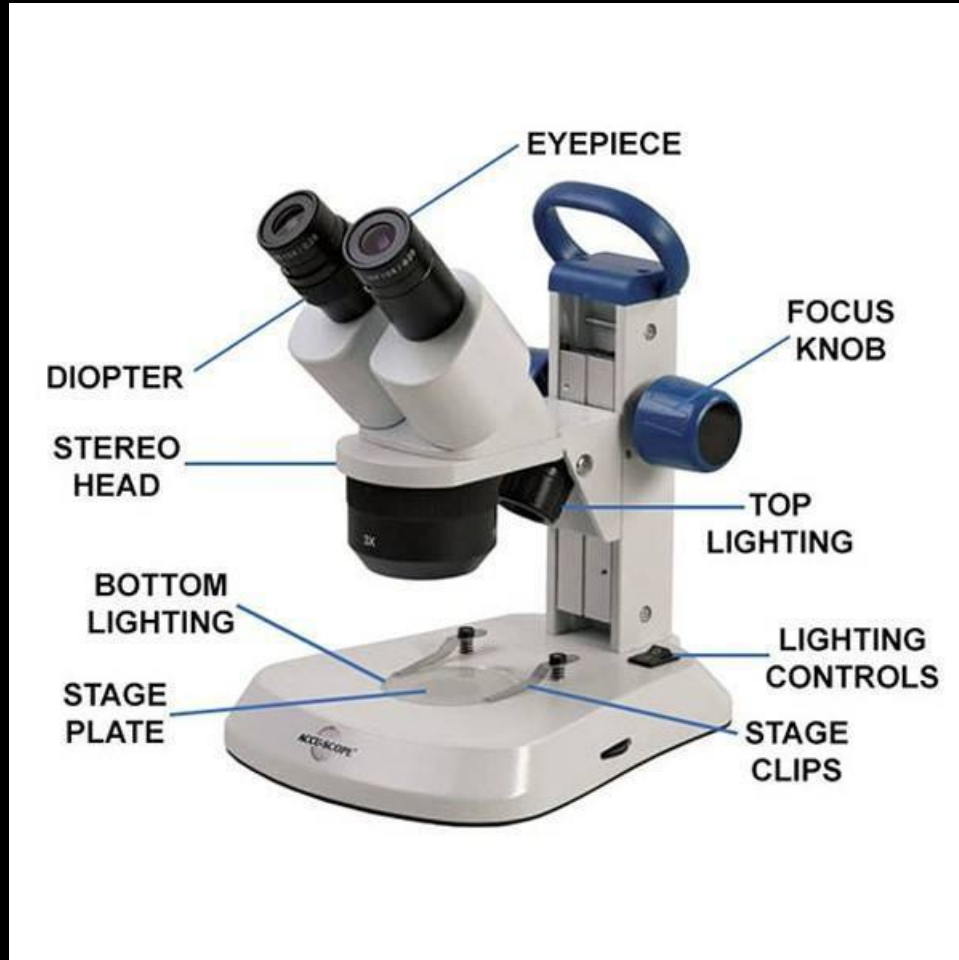


<https://www.thermofisher.com/blog/atomic-resolution/seeing-with-electrons-the-anatomy-of-an-electron-microscope/>

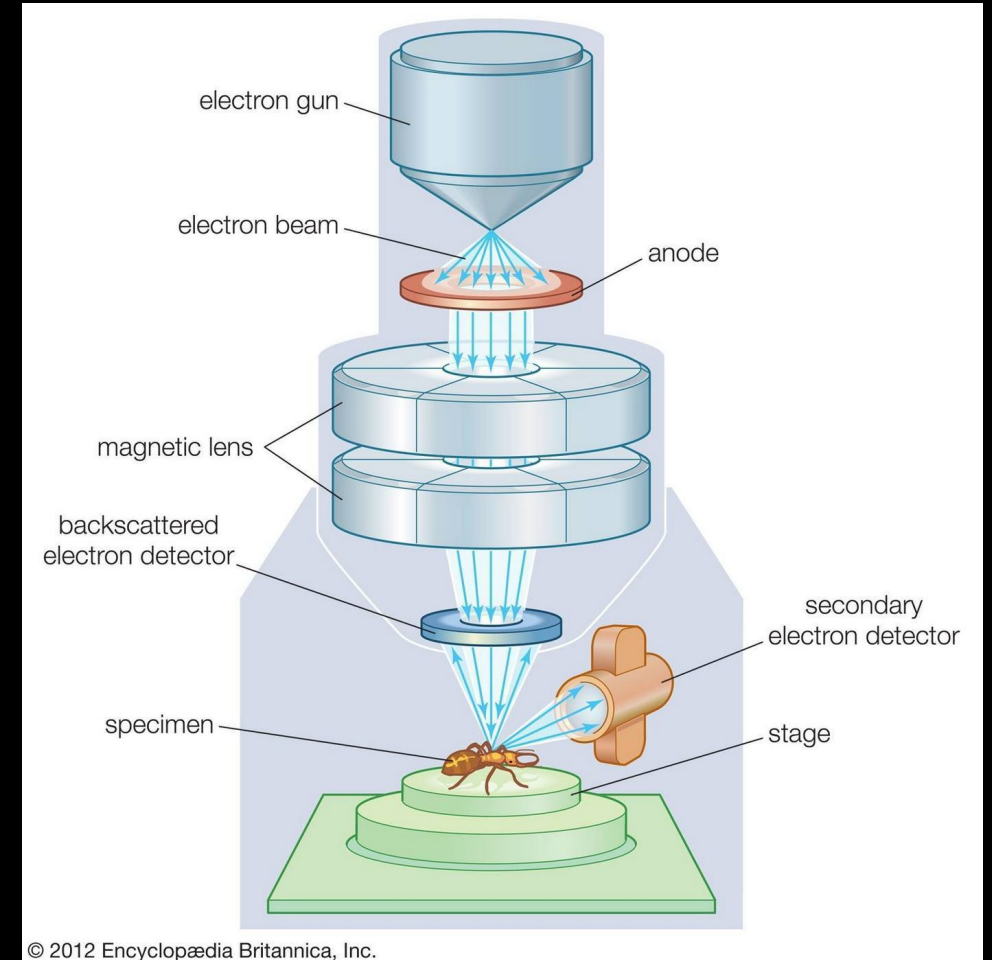


A TEM micrographs of a common single-celled green algae (Chlamydomonas).

Though the analogy is not as solid as that for the TEM, an SEM is analogous to a stereomicroscope

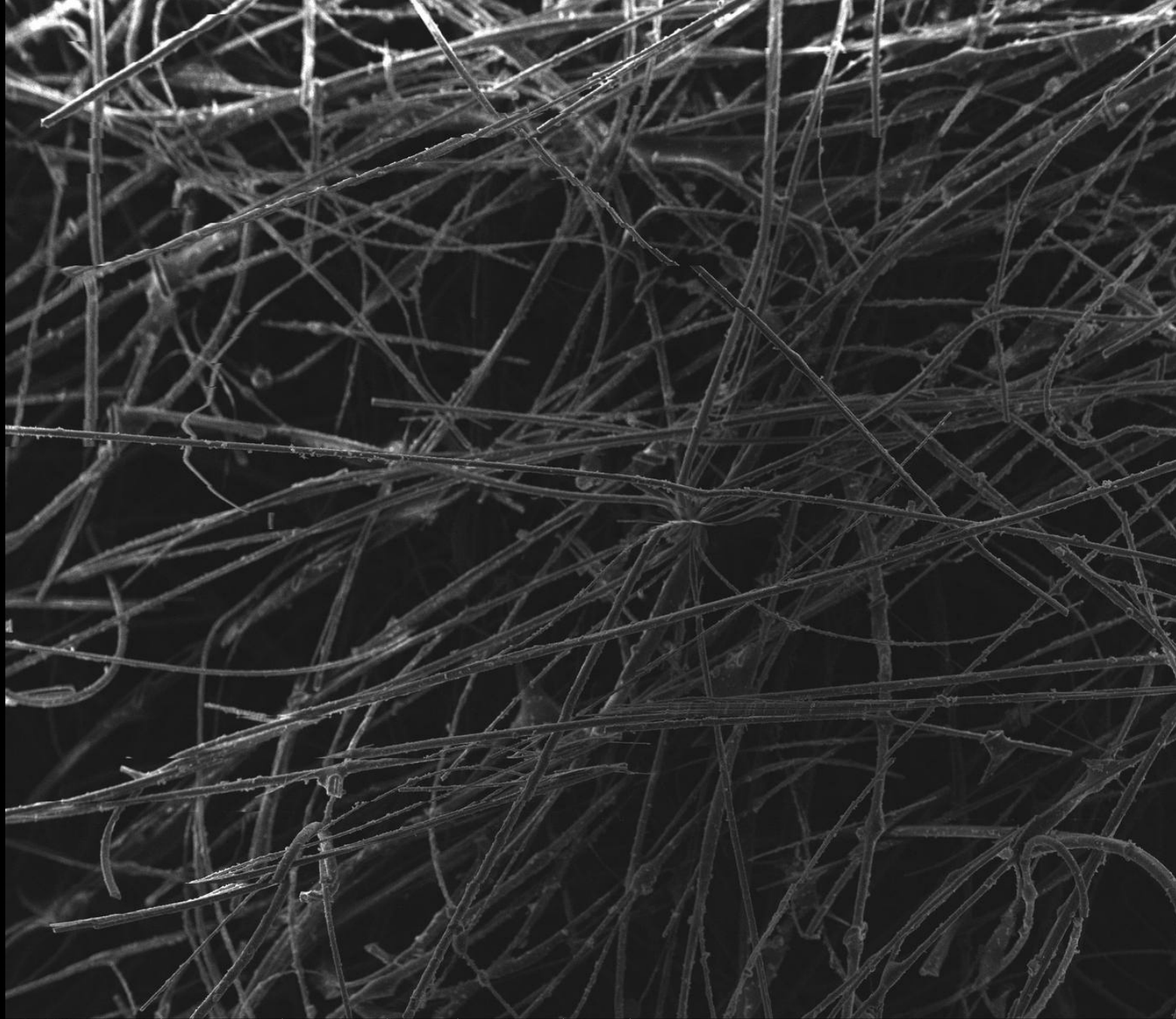


<https://microscopeinternational.com/what-is-a-stereo-microscope/>



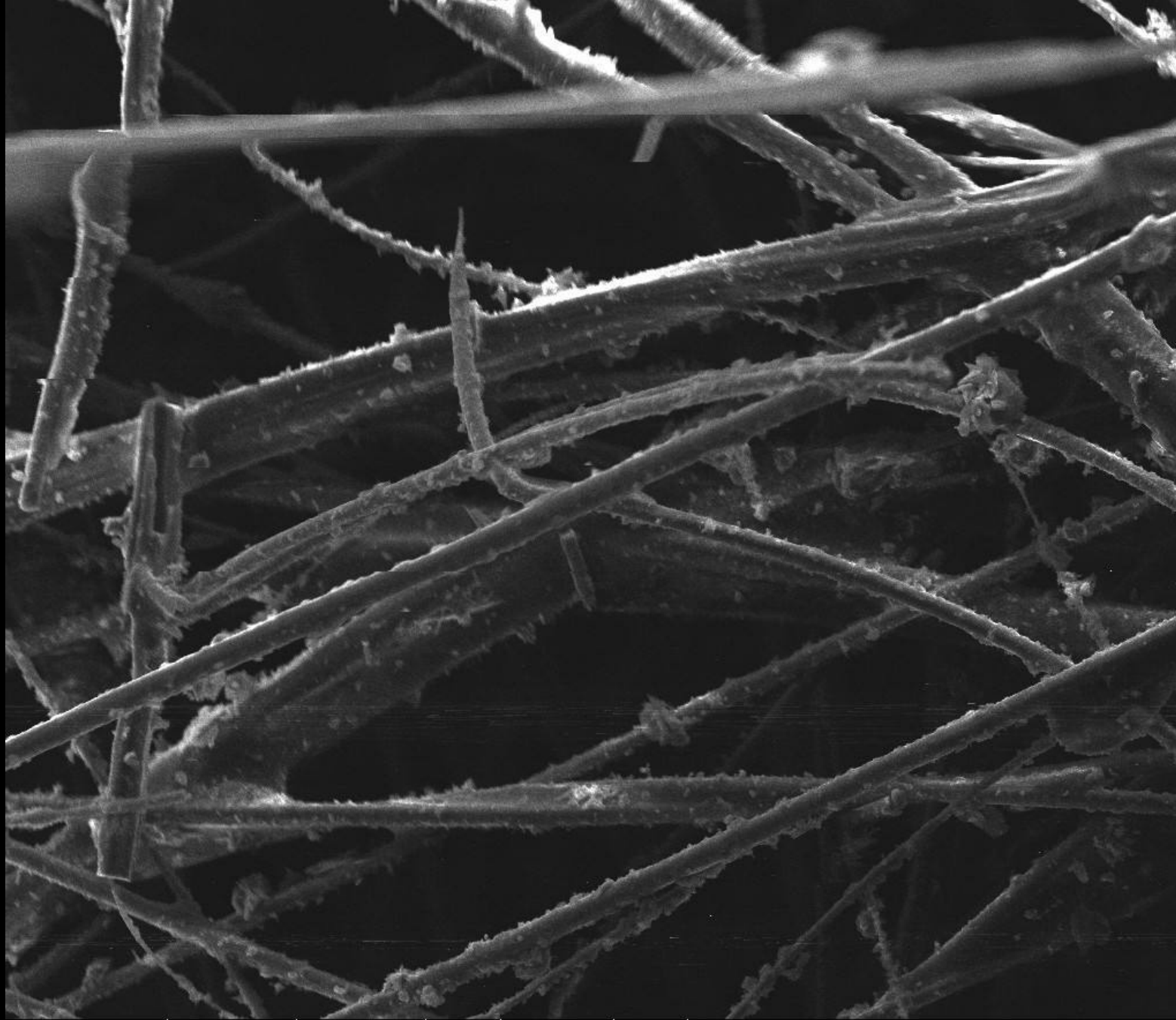
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<https://www.britannica.com/technology/scanning-electron-microscope>



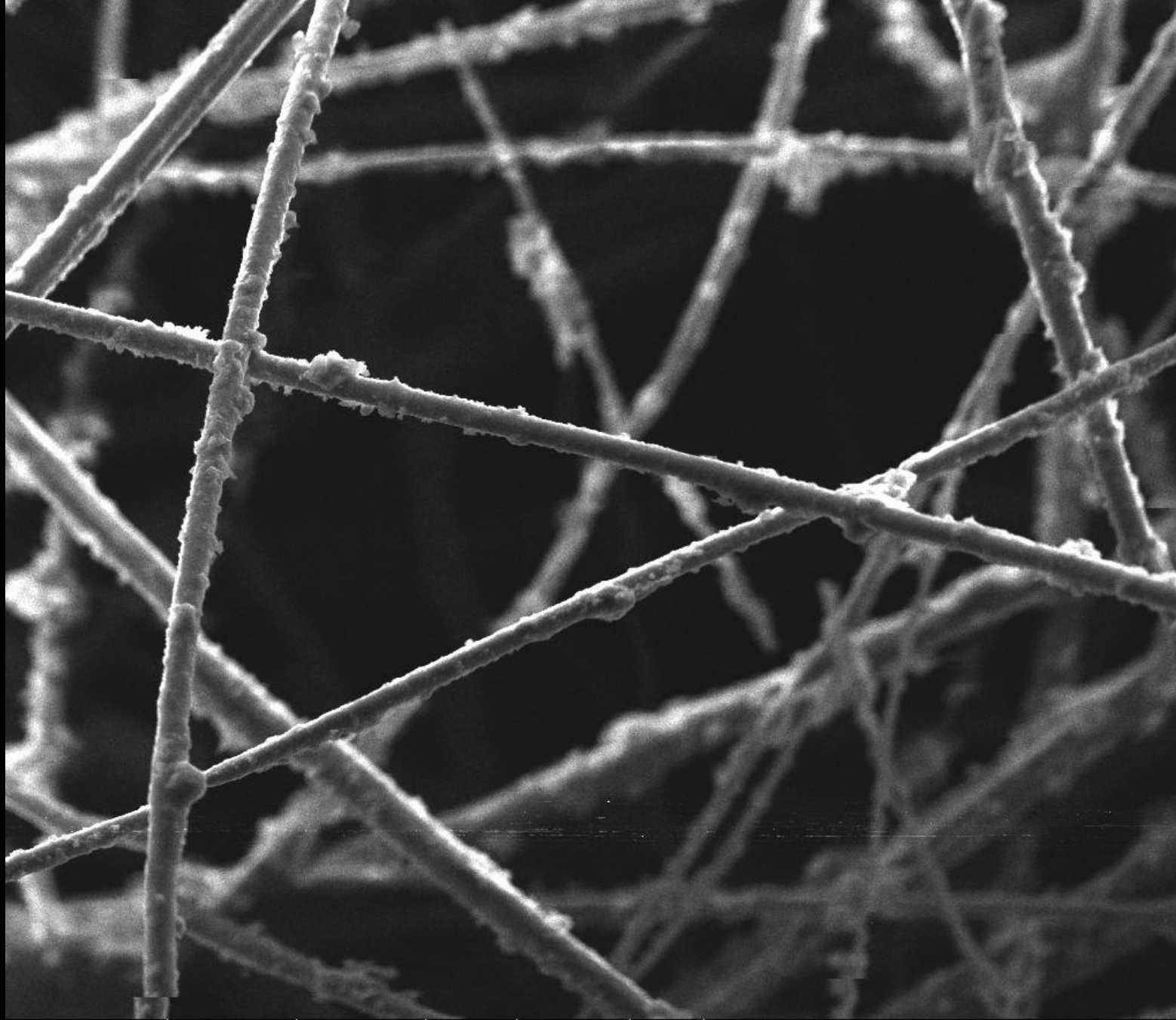
9/13/2023	HFW	WD	Mag	HV	Spot
9:26:11 AM	1.42 mm	9.8 mm	180x	20.0 kV	3.0

500.0µm



9/13/2023	HFW	WD	Mag	HV	Spot
9:28:00 AM	0.32 mm	10.3 mm	800x	20.0 kV	3.0

100.0µm

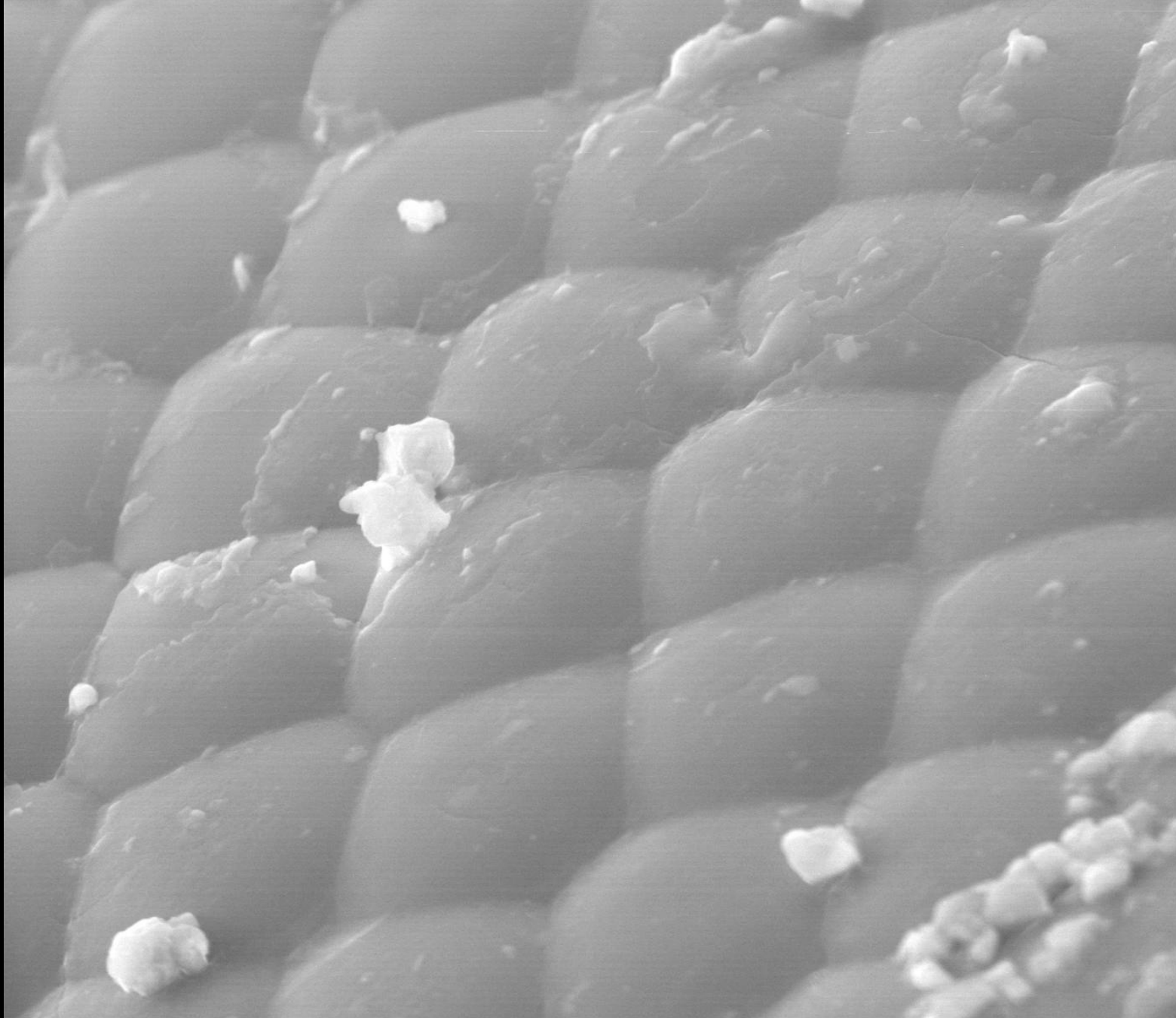


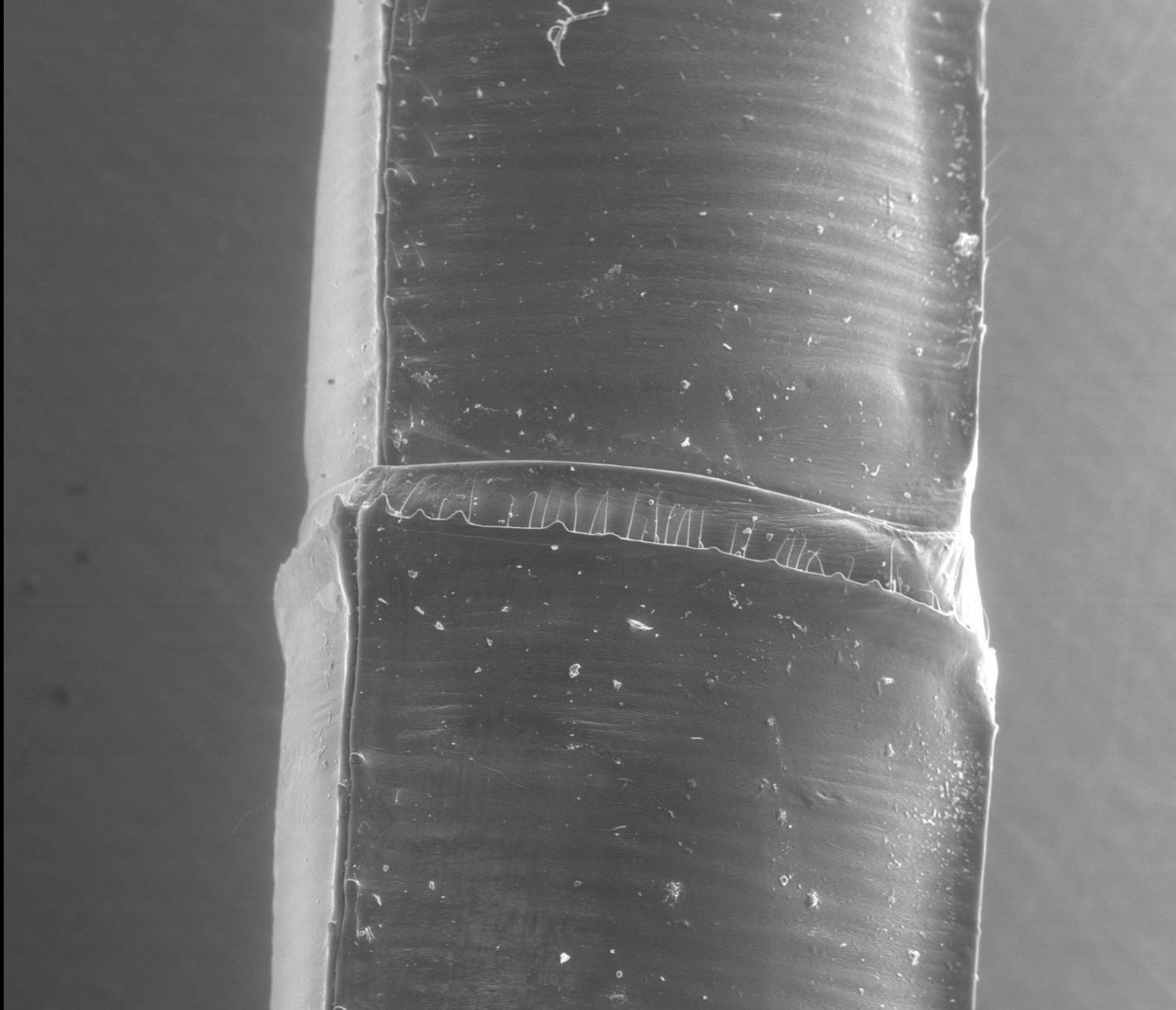
9/13/2023	HFW	WD	Mag	HV	Spot	100.0μm
9:31:23 AM	0.26 mm	10.0 mm	1000x	20.0 kV	3.0	



HV	Spot	Mag	Sig	WD	Pressure
20.0 kV	3.3	400x	SE	8.51 mm	130.0 Pa

—200 μ m—
Ant Head





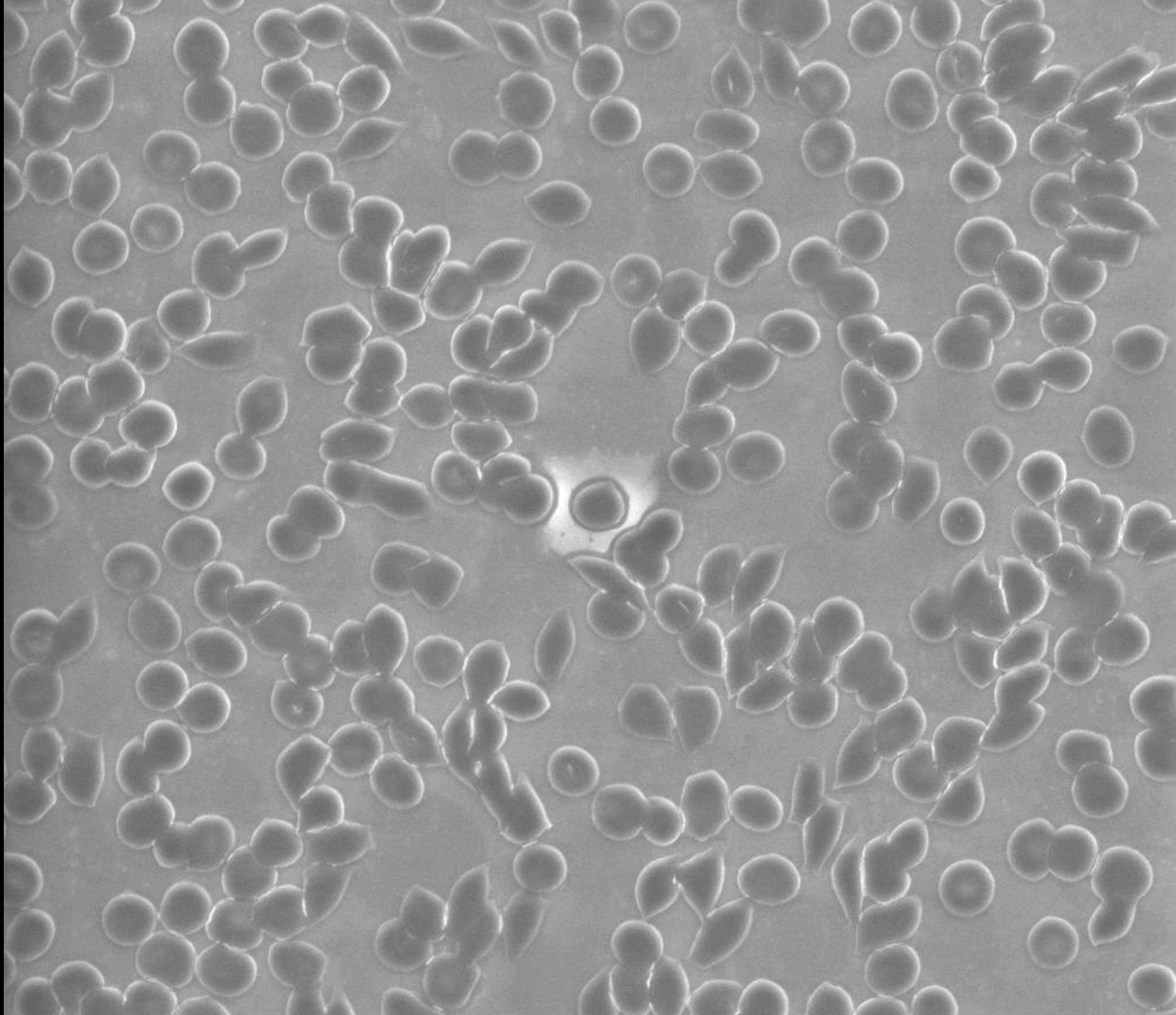
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—500 μ m—
Bug's Knee



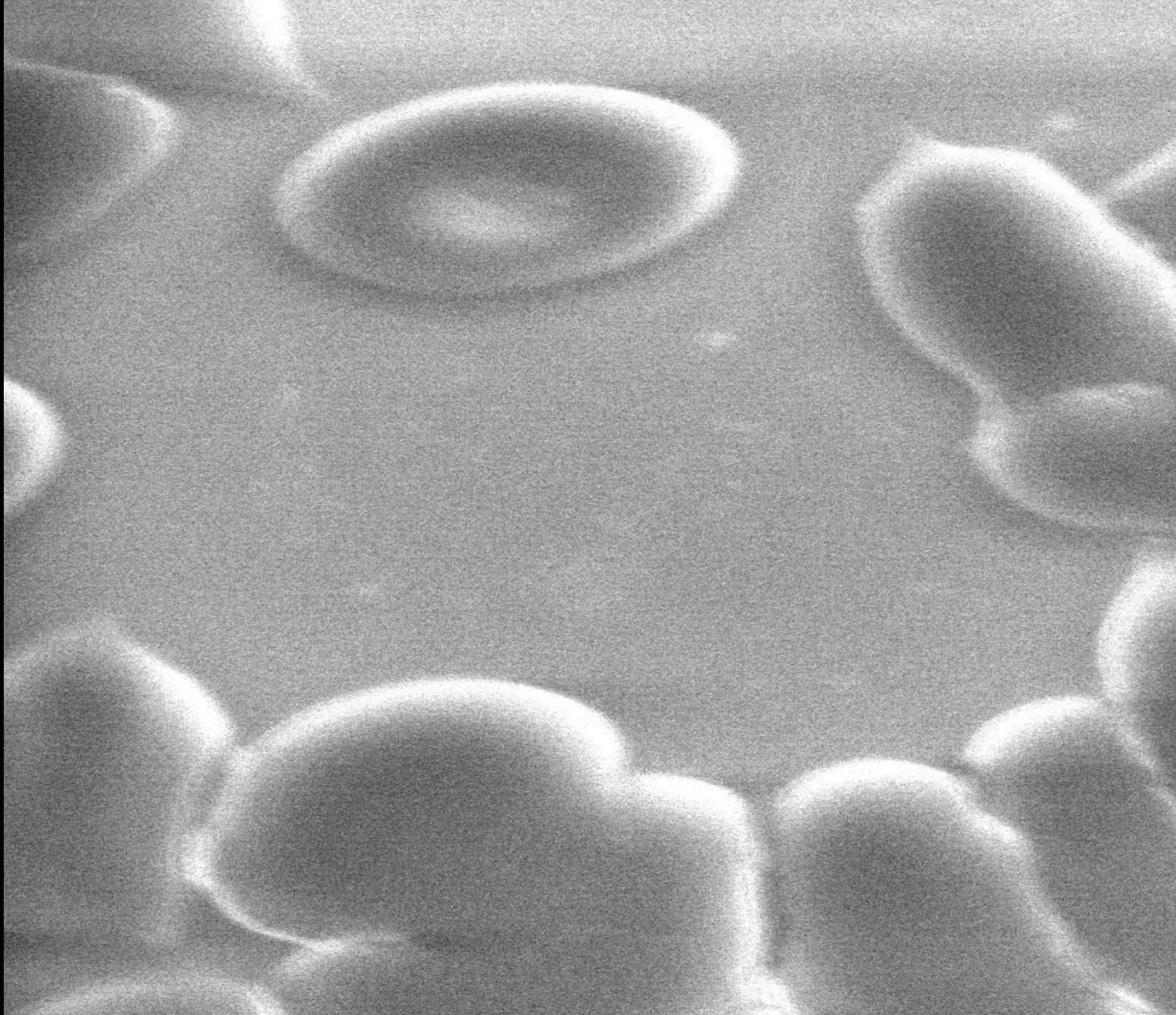
HV	Spot	Mag	Sig	WD	Pressure
20.0 kV	4.0	400x	SE	6.04 mm	130.0 Pa

—200 µm—
Gary's Flea



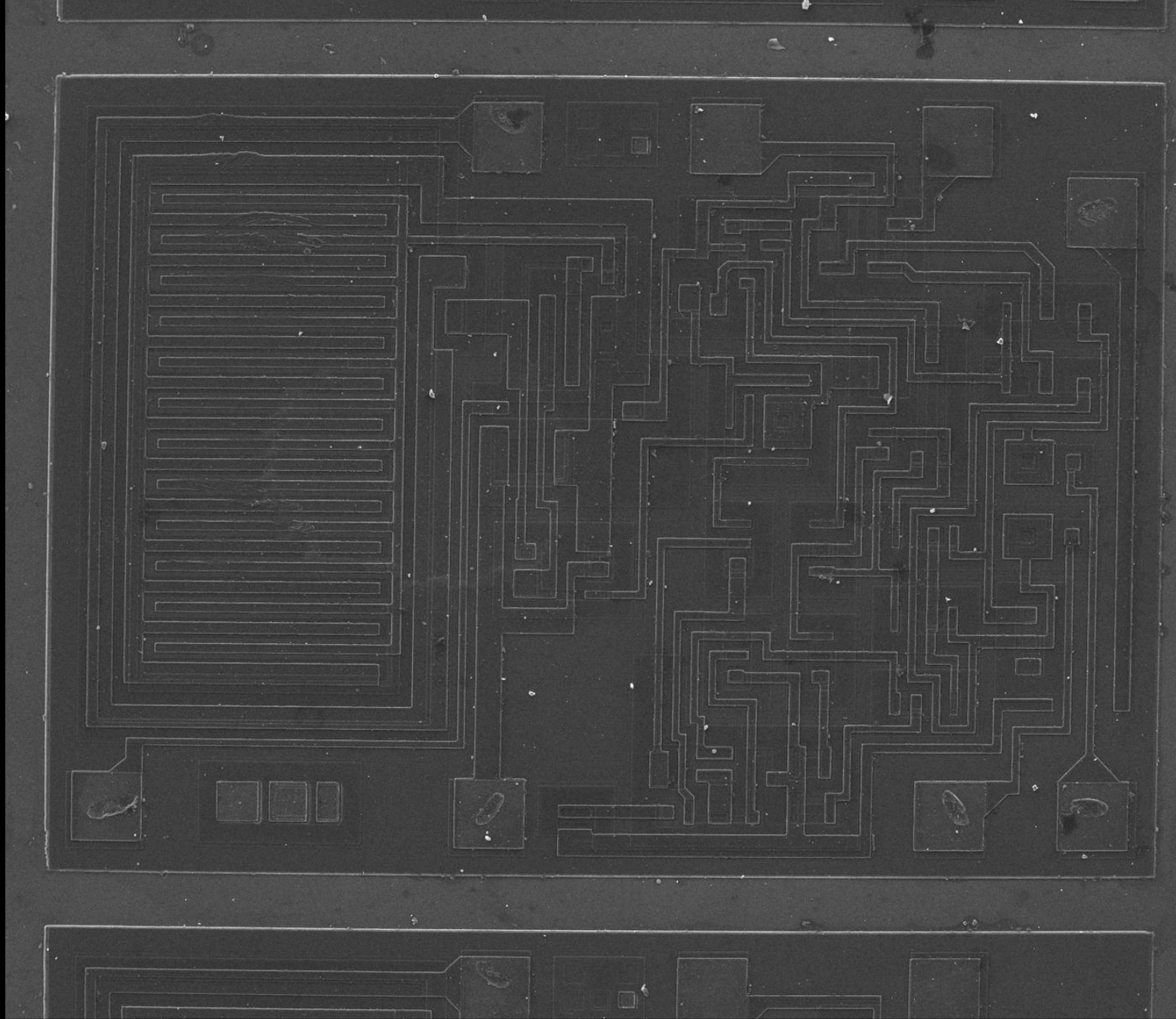
HV	Spot	Mag	Sig	WD	05/14/10	Pressure
12.5 kV	4.0	1485x	SE	10.63 mm	17:16:46	90.0 Pa

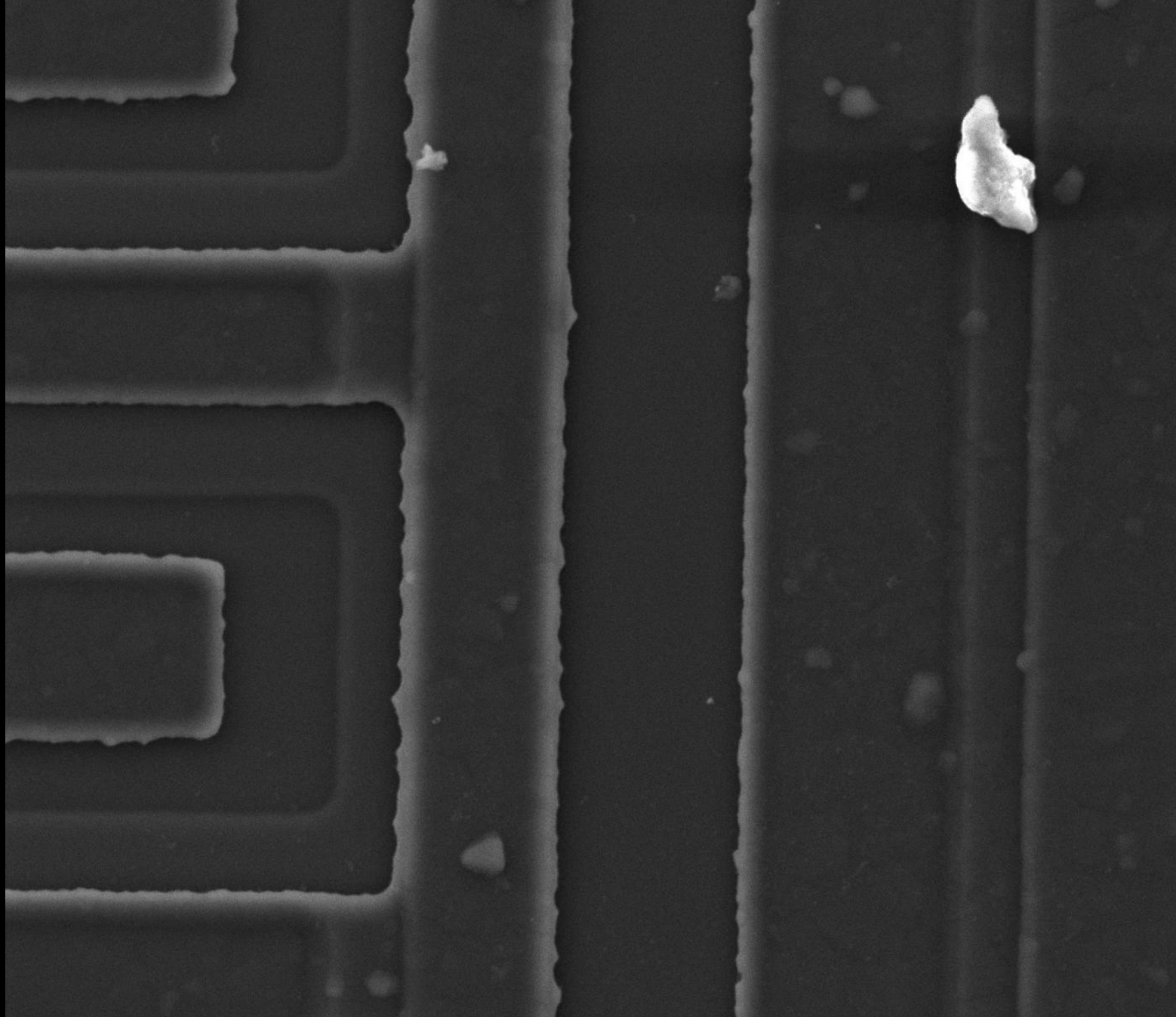
—20 µm—

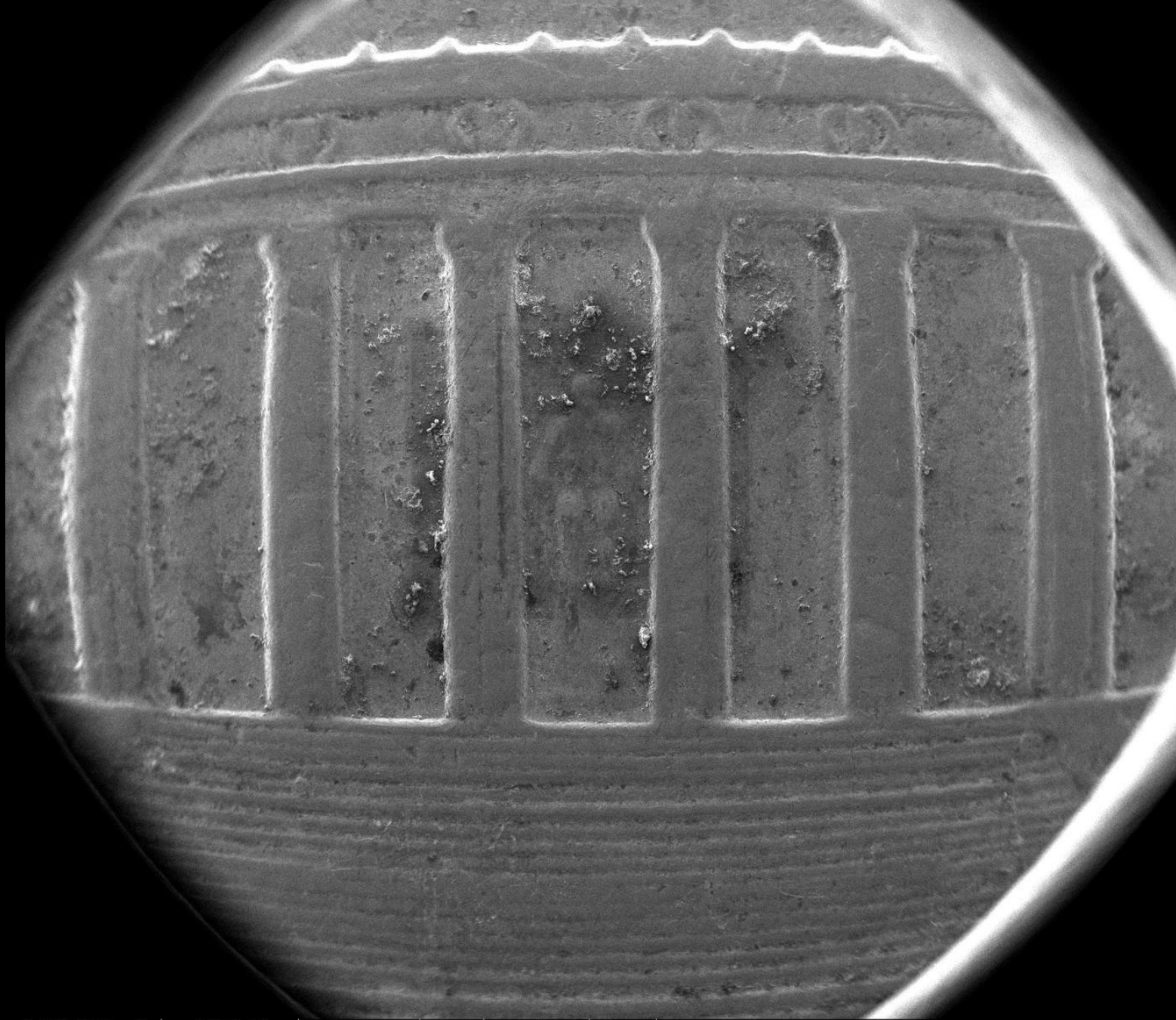


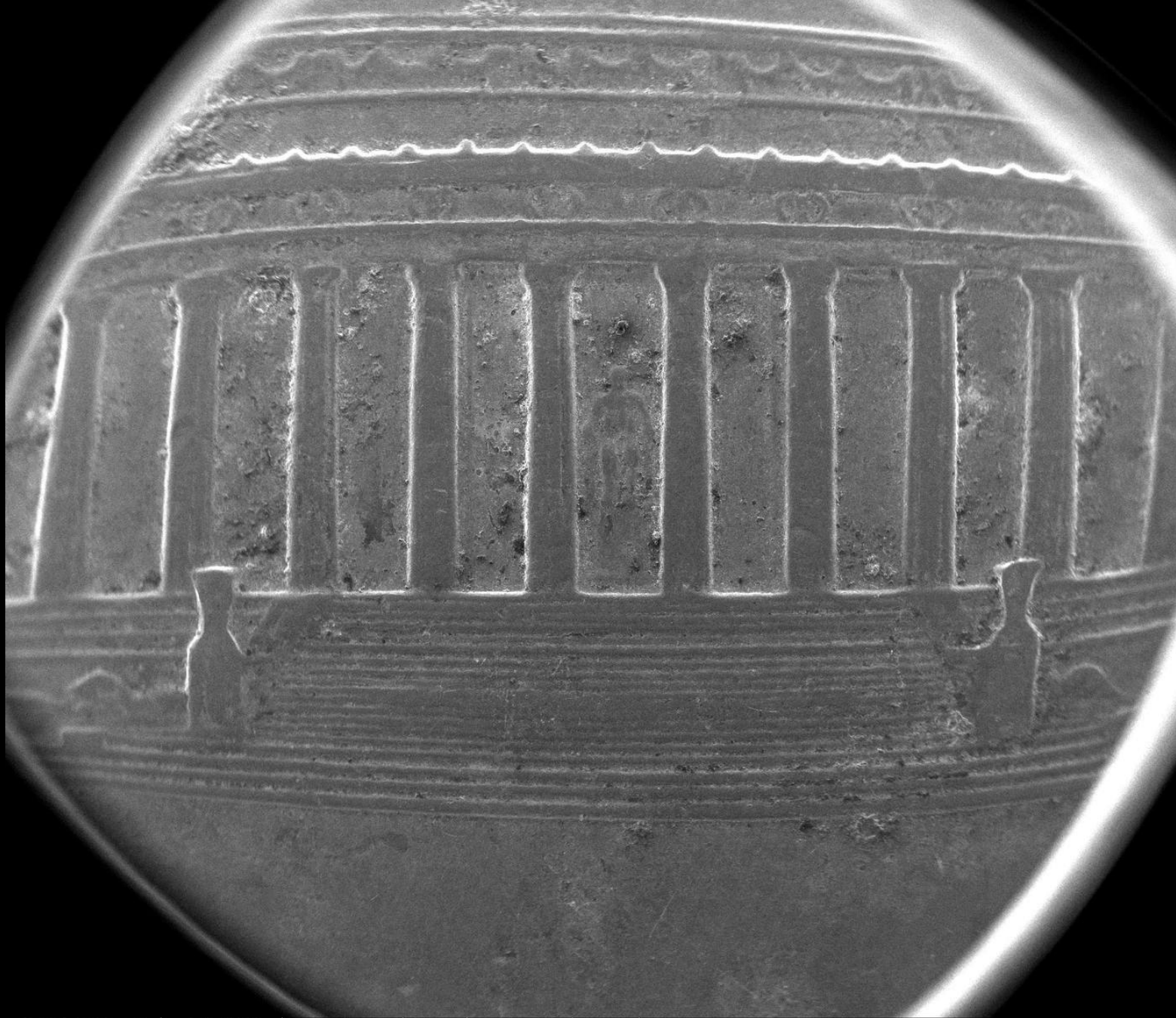
HV	Spot	Mag	Sig	WD	05/14/10	Pressure
12.5 kV	3.5	12144x	SE	10.69 mm	17:05:29	130.0 Pa

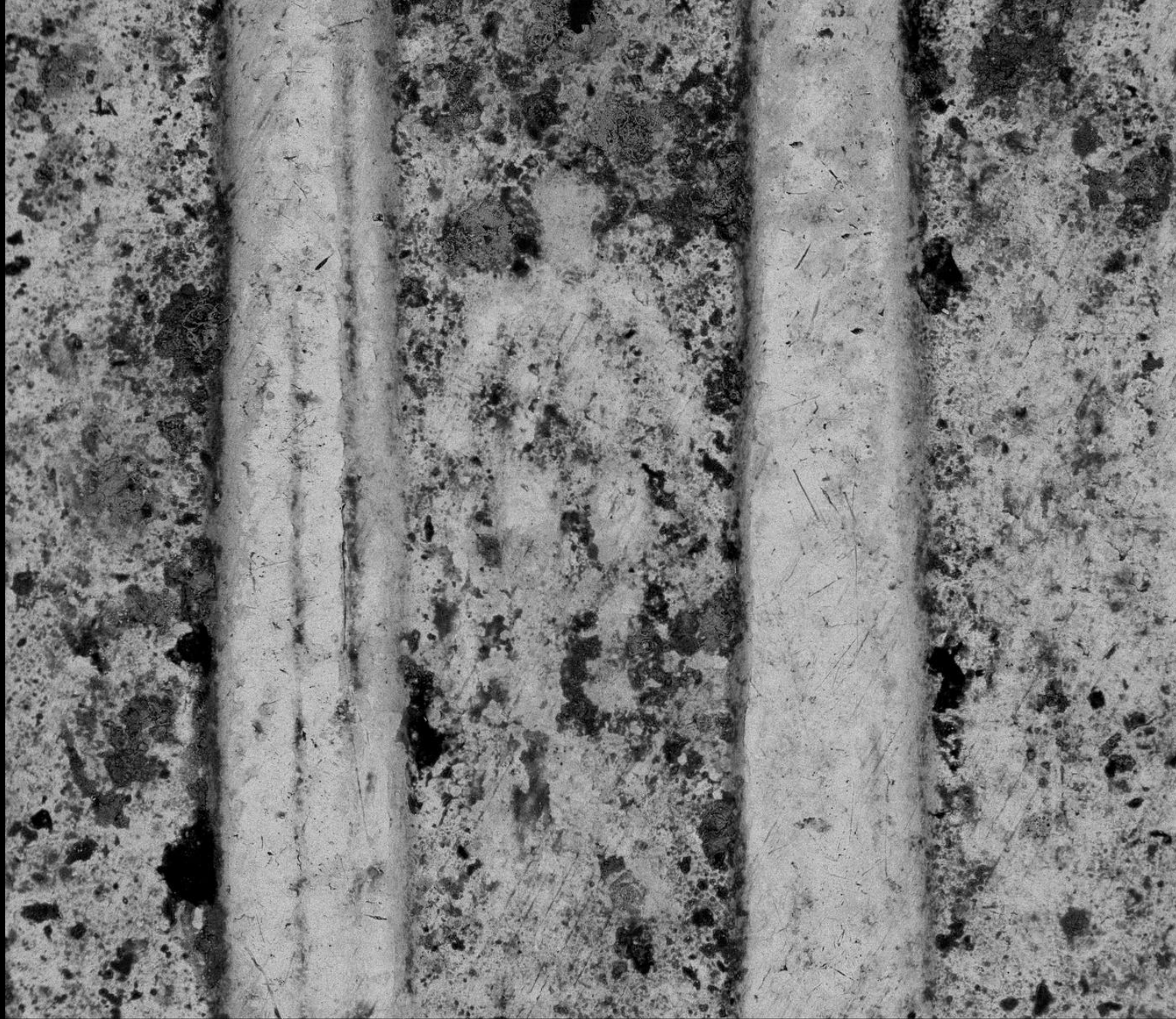
—5 μm—





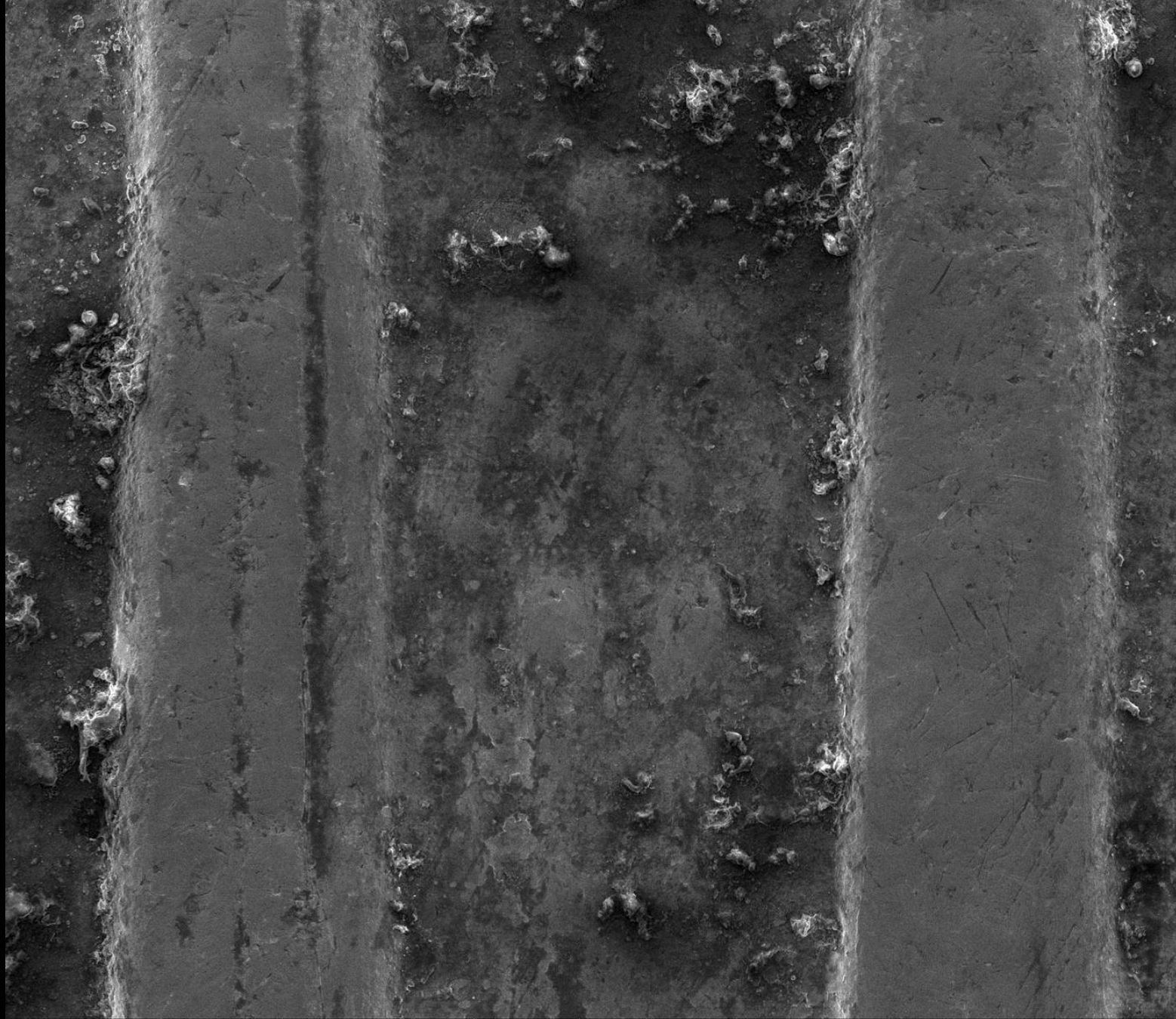






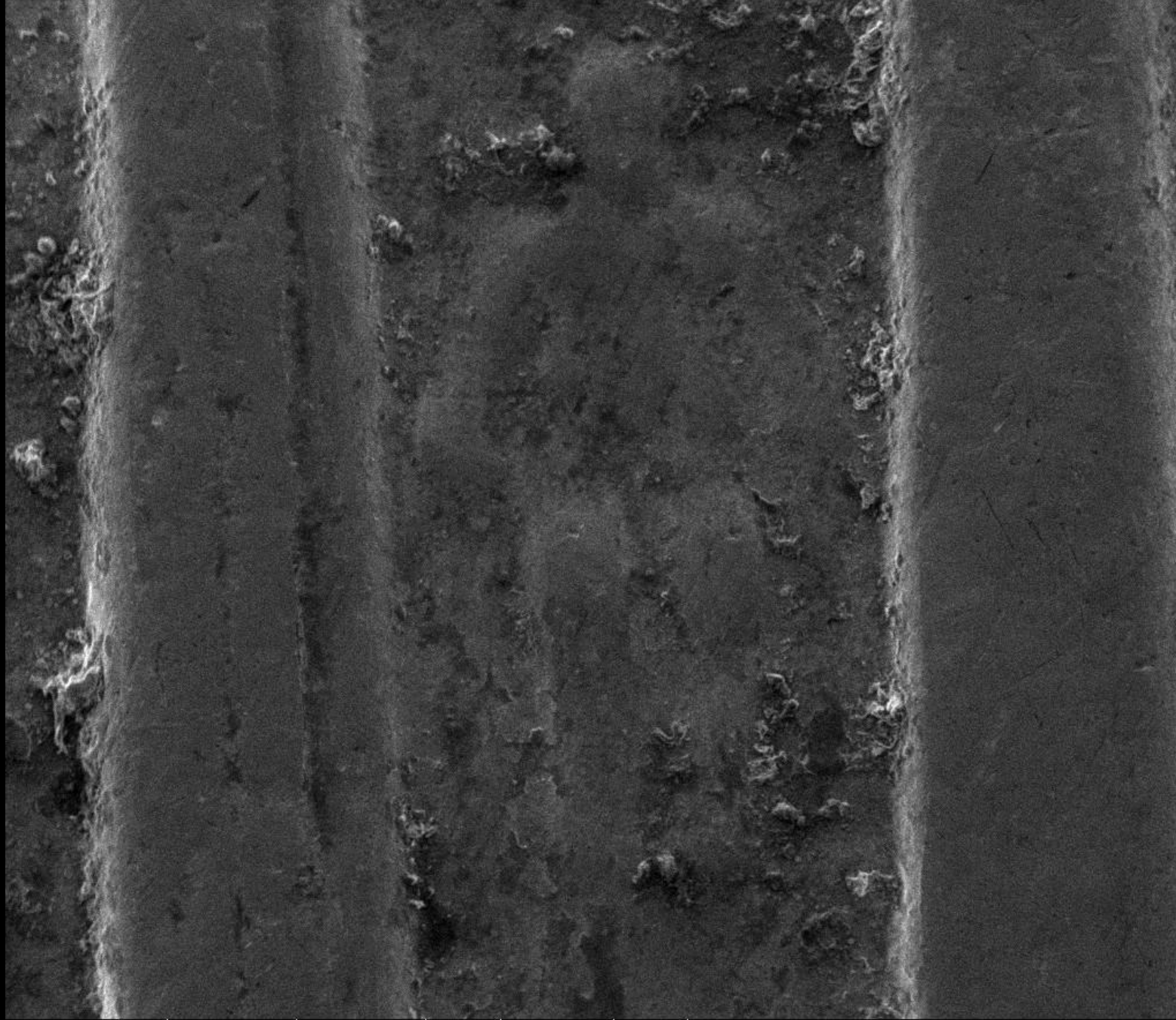
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2:01:28 PM	2.58 mm	10.6 mm	99x	20.0 kV	3.0

1.0mm

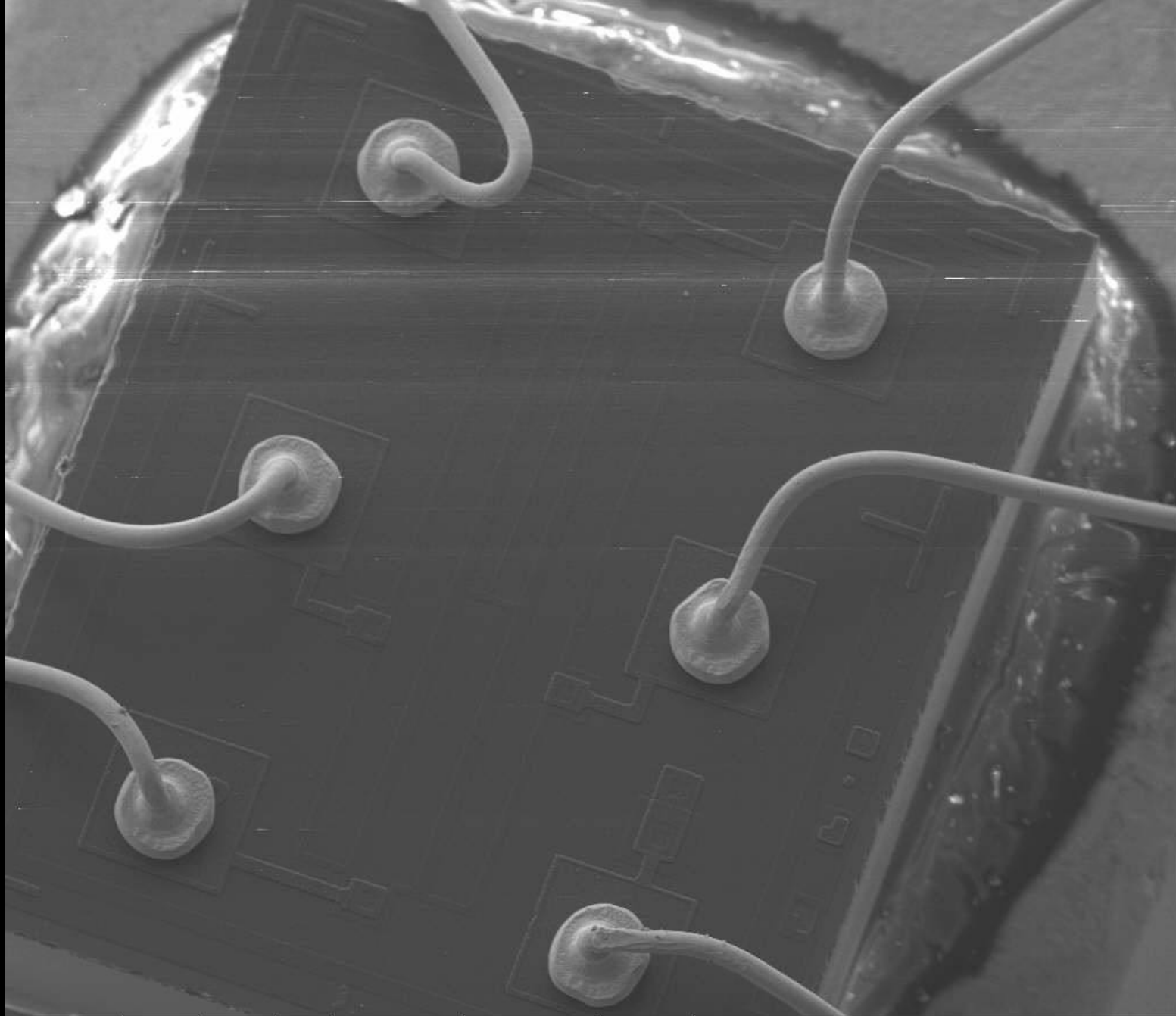


9/27/2023	HFW	WD	Mag	HV	Spot
1:57:13 PM	1.85 mm	22.5 mm	138x	20.0 kV	3.0

500.0µm

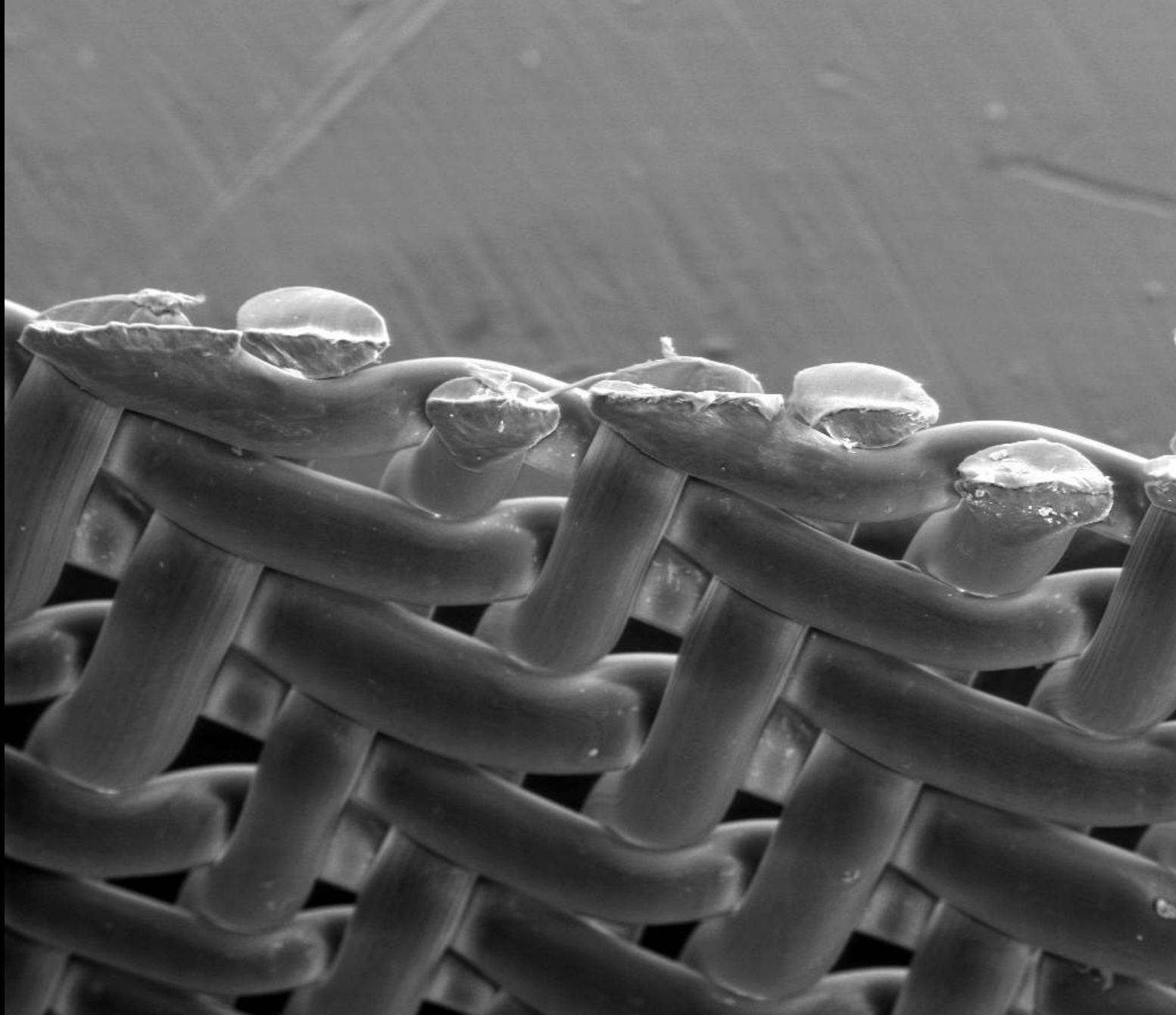


9/18/2023	HFW	WD	Mag	HV	Spot	500.0μm
3:54:51 PM	1.56 mm	10.3 mm	165x	20.0 kV	3.0	



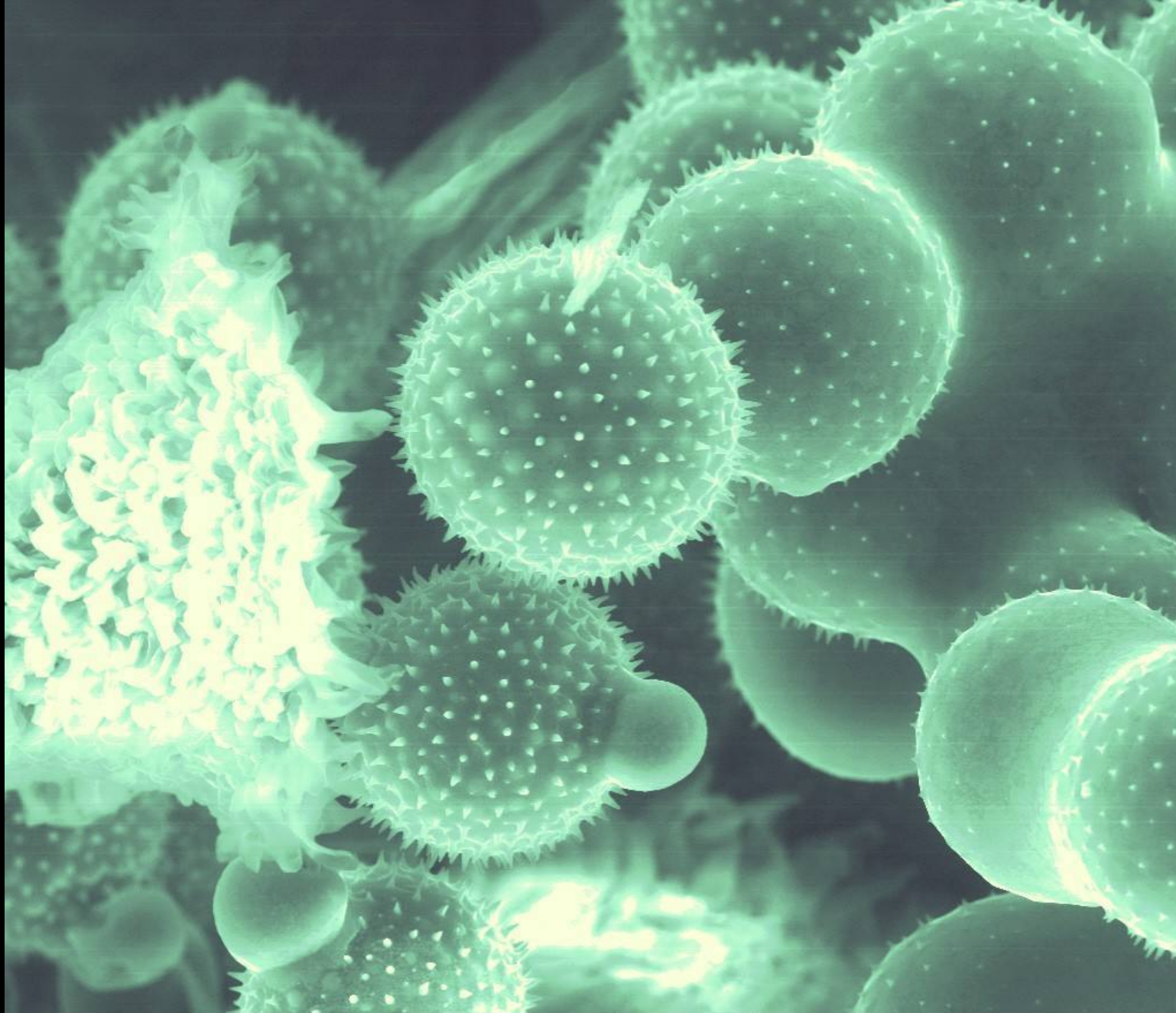
HV	Spot	Mag	Sig	WD	Pressure	Tilt
20.0 kV	3.5	208x	SE	9.65 mm	---	13.3 °

—200 μm—



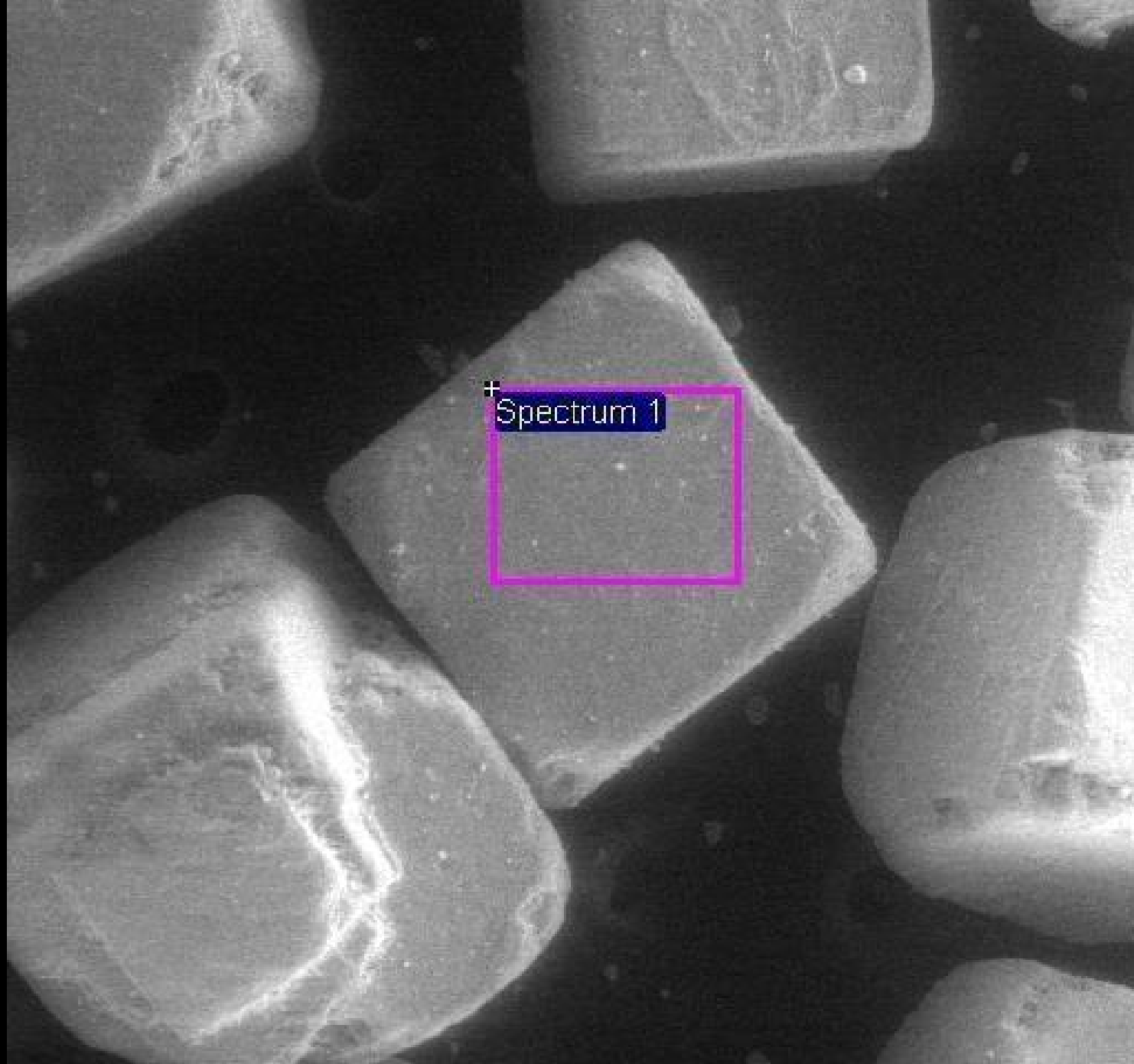
HV	Spot	Mag	Sig	WD	Tilt	03/16/09
20.0 kV	4.0	240x	SE	12.85 mm	44.4 °	14:12:07

—200 μm—



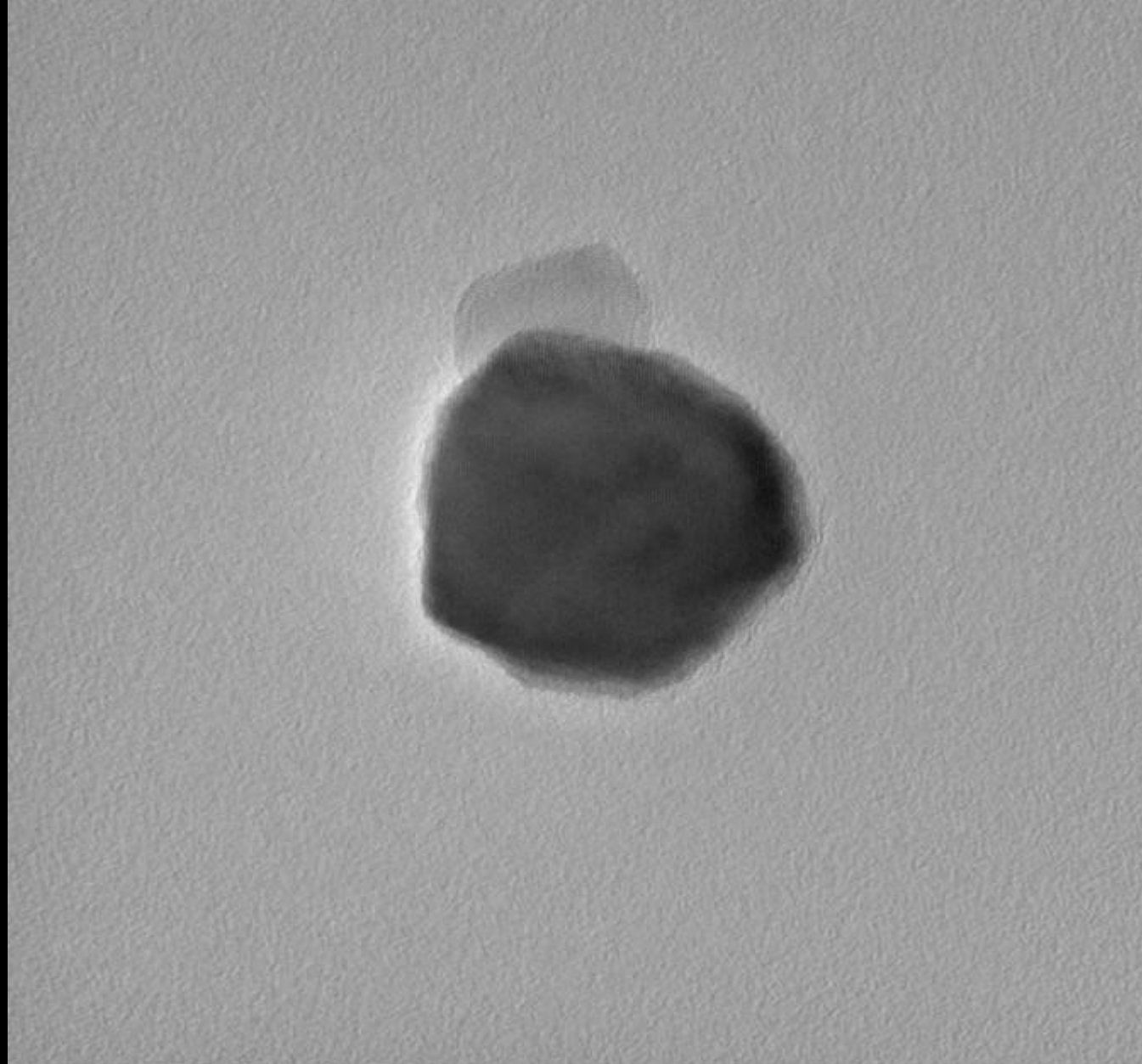
HV	Mag	WD	08/07/02	Pressure	Tilt	Sig
30.0 kV	800x	17.51 mm	12:04:30	73.3 Pa	2.6 °	SE

—50 μm—



Spectrum 1



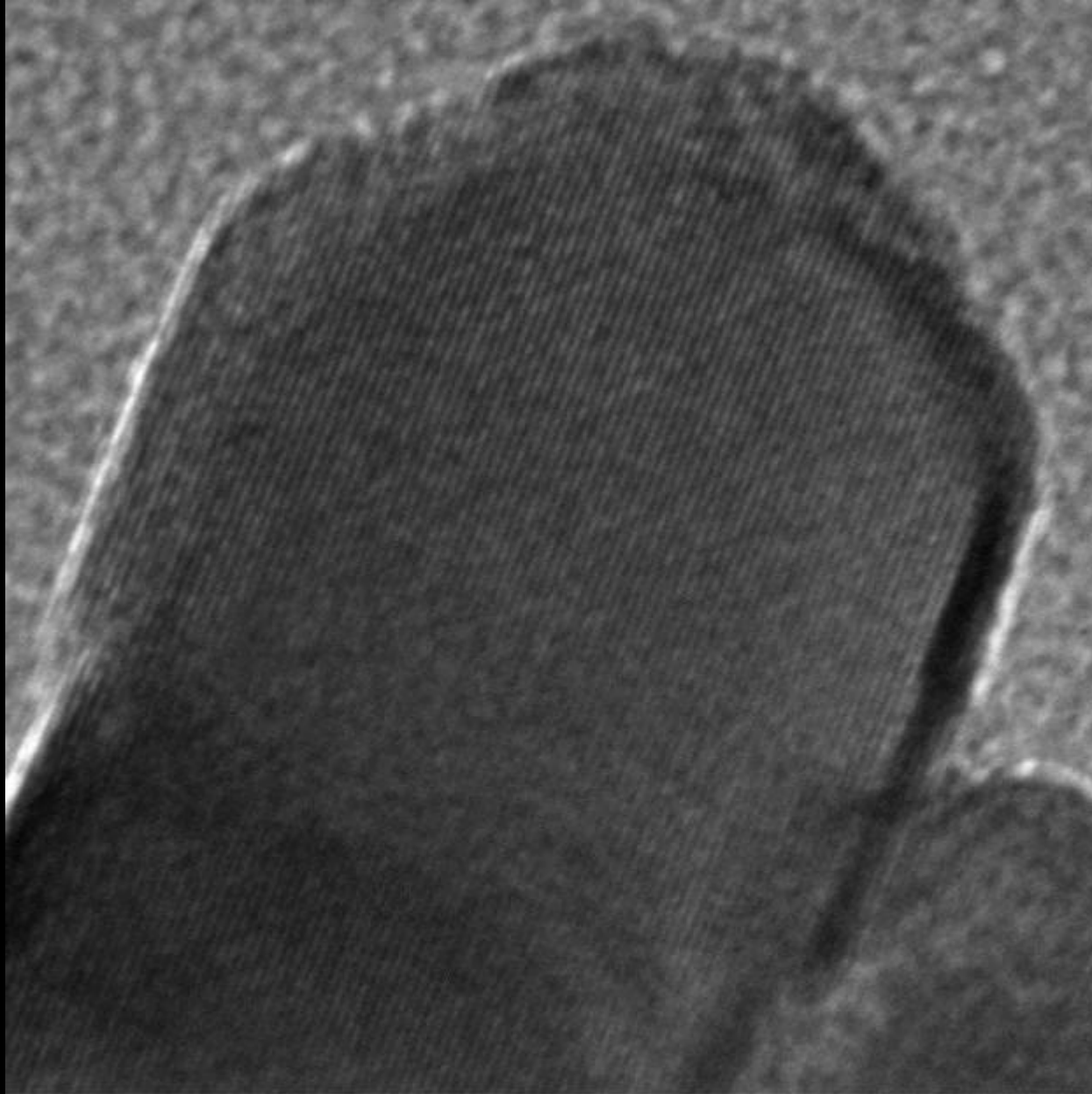


8Ce2Zr_N-2_004
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5 nm
HV=200.0kV
Direct Mag: 1500000x
Kettering University

8Ce2Zr_N-2_005
11:45:28 5/18/2017

0.2 (1/Å)
HV=200.0kV
Cam Len: 1.2000 m
Kettering University



LCMO10-4.tif

Print Mag: 3970000x @ 7.0 in

5 nm

HV=200.0kV

Direct Mag: 400000x

AMT Camera System